# Action Archive

## for BigFix Platform

## Administrator Guide

| | |
|---|---|
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| **Platform** | Ubuntu Server 24.04 / 26.04 LTS |
| **Author** | Mike Consuegra |

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# Table of Contents

# 1. Introduction

## 1.1 Purpose of This Guide

This guide provides comprehensive reference and procedural documentation for administrators responsible for deploying, configuring, and maintaining the Action Archive for BigFix Platform application. It covers all administrative functions available through the web interface as well as server-side operations performed via the Linux command line.

## 1.2 Intended Audience

This guide is written for:

- BigFix administrators who manage day-to-day operations and will serve as the primary users of the application
- IT operations staff who maintain the underlying server infrastructure
- Security and compliance personnel who require audit visibility into BigFix action history

Readers are assumed to be comfortable with BigFix concepts (operators, fixlets, actions, sites), Linux system administration, and general networking. BigFix itself is not described in this guide.

## 1.3 Document Conventions

| Convention | Meaning |
|---|---|
| ℹ️ NOTE | Additional context, clarification, or technical background |
| ▶️ TIP | A helpful suggestion or best practice |
| ❗ IMPORTANT | A critical requirement that must be understood before proceeding |
| ⚠️ WARNING | A condition or action that may result in data loss or a security risk |
| ✘ CRITICAL | Severe risk; take immediate action before proceeding |
| Monospace / COMMAND | Terminal commands, file paths, configuration values, and code samples |

## 1.4 Application Overview

Action Archive for BigFix Platform is an community-created web application that captures, preserves, and presents stopped and expired actions visible in the BigFix Console. Because the BigFix Console does not retain detailed records of actions once deleted, Action Archive fills this gap by periodically querying the BigFix REST API and storing full action metadata in a local PostgreSQL database so they can safely be deleted from the console without losing historical visibility.

Key value propositions:

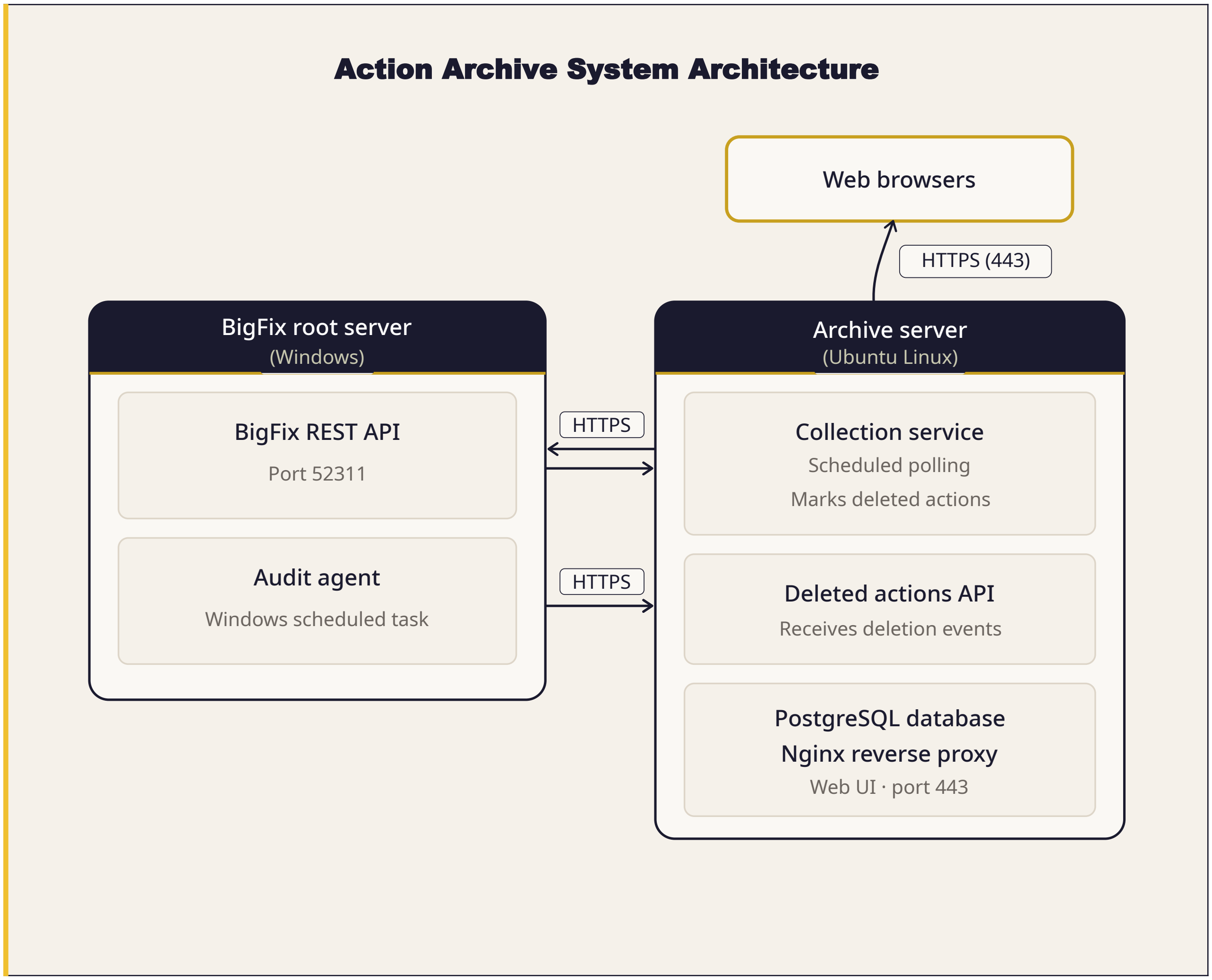
- Compliance and audit readiness: a permanent, searchable record of every BigFix action and its per-device results
- Operational visibility: instant answers to "Who deployed this patch?" and "Which devices failed?"
- Deletion tracking: the Audit Agent captures action deletions that would otherwise leave no record in the BigFix Console.
- Role-based access: separate admin, auditor, and viewer roles enforce least-privilege access
- Export capability: action lists and audit logs exportable to CSV and PDF for compliance reporting

> ℹ️ **NOTE** Action Archive collects expired and stopped actions only. It is not a real-time monitoring tool and does not interfere with BigFix operations in any way.

# 2. Architecture Overview

## 2.1 System Architecture Diagram

The following diagram illustrates the major components of the Action Archive deployment and the data flows between them.



## 2.2 Component Descriptions

| Component | Technology | Role |
|---|---|---|
| Archive Server | Ubuntu Server 22.04 / 24.04 LTS | Host VM for all application services |
| Reverse Proxy | Nginx | Terminates TLS, enforces security headers, proxies to FastAPI on localhost:5000 |
| Application Server | Python 3.12 / FastAPI / Uvicorn | Handles all API requests, collection logic, authentication, and business rules |
| Database | PostgreSQL 16 | Stores archived actions, users, settings, audit logs, and collection history |
| BigFix Root Server | HCL BigFix Platform | Source of action data; queried via REST API (port 52311) and Session Relevance |
| Audit Agent | Python / Windows Scheduled Task | Monitors server_audit.log on BigFix server; reports action deletions to Archive API |
| Browser Clients | Google Chrome, Mozilla Firefox, Microsoft Edge | Access the Action Archive web UI over HTTPS (port 443) |

## 2.3 Network Requirements

| Port | Protocol | Direction | Purpose |
|---|---|---|---|
| 443 | TCP / HTTPS | Inbound to Archive Server | Web UI and API access from browsers and the Audit Agent |
| 52311 | TCP / HTTPS | Outbound from Archive Server | BigFix REST API and Session Relevance queries |
| 5000 | TCP / HTTP | Localhost only | Internal: Nginx → FastAPI (not exposed externally) |
| 5432 | TCP | Localhost only | Internal: FastAPI → PostgreSQL (not exposed externally) |
| 25/587/465 | TCP / SMTP | Outbound from Archive Server | Email notifications and password reset (if SMTP configured) |

> ❗ **IMPORTANT** Ports 5000 and 5432 are bound to localhost only and must never be exposed externally. All external access occurs exclusively through Nginx on port 443.

## 2.4 Deployment Model

Action Archive is a single-instance application. Only one BigFix Root Server is supported for each Archive Server. There is no clustering or high-availability configuration. Redundancy is achieved through regular database backups and VM-level snapshots of the host.

The application is designed to support several concurrent users. Storage growth is estimated at approximately 15 MB per 1,000 actions.

# 3. Getting Started: First Login

## 3.1 Accessing the Web Interface

Open a web browser and navigate to:

COMMAND
```
https://<server-ip>
  or
https://<server-hostname>
```

## 3.2 Self-Signed Certificate Warning

By default, Action Archive uses a self-signed SSL certificate. Your browser will display a security warning on first access. This is expected behavior in a standard deployment.

To proceed through the warning:

1. Click Advanced or More Information.
2. Click Proceed to <address> or Accept the Risk and Continue.
3. The login page will load normally. The warning will not reappear in the same browser.

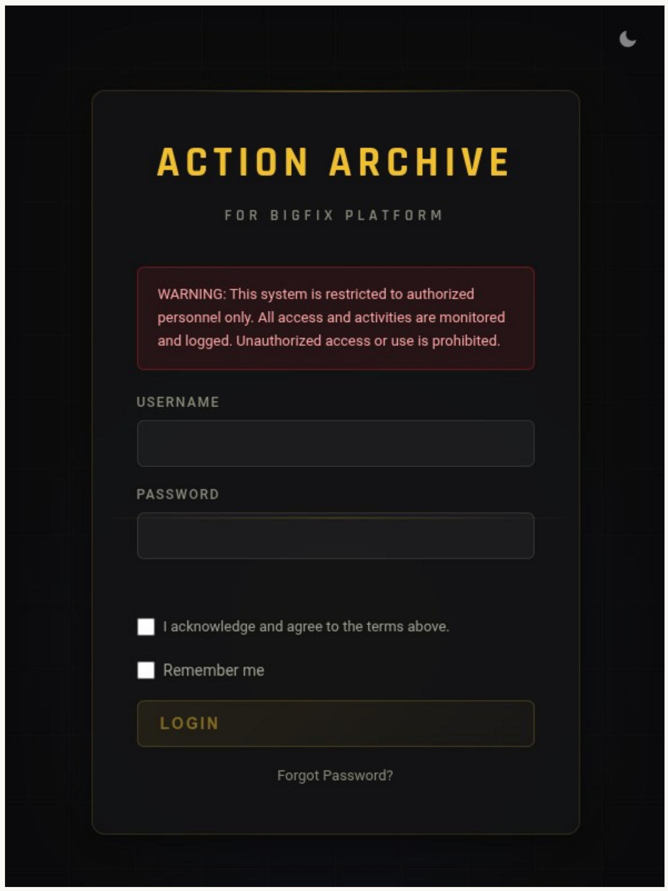
> ℹ️ **NOTE** To eliminate the browser warning permanently, replace the self-signed certificate with one signed by your internal Certificate Authority. See Section 9.5 (Certificates) for instructions.

## 3.3 Login Page

The login page provides the following interactive features:

- Theme toggle: Switch between Dark and Light themes. Your selection persists across sessions.
- **[If Configured]:** Use Consent checkbox: Must be checked before login is permitted. Confirms acknowledgment of acceptable-use terms.
- **[If Configured]:** Remember Me: Extends the session token lifetime.

Enter the administrator credentials supplied during installation, then click Sign In.



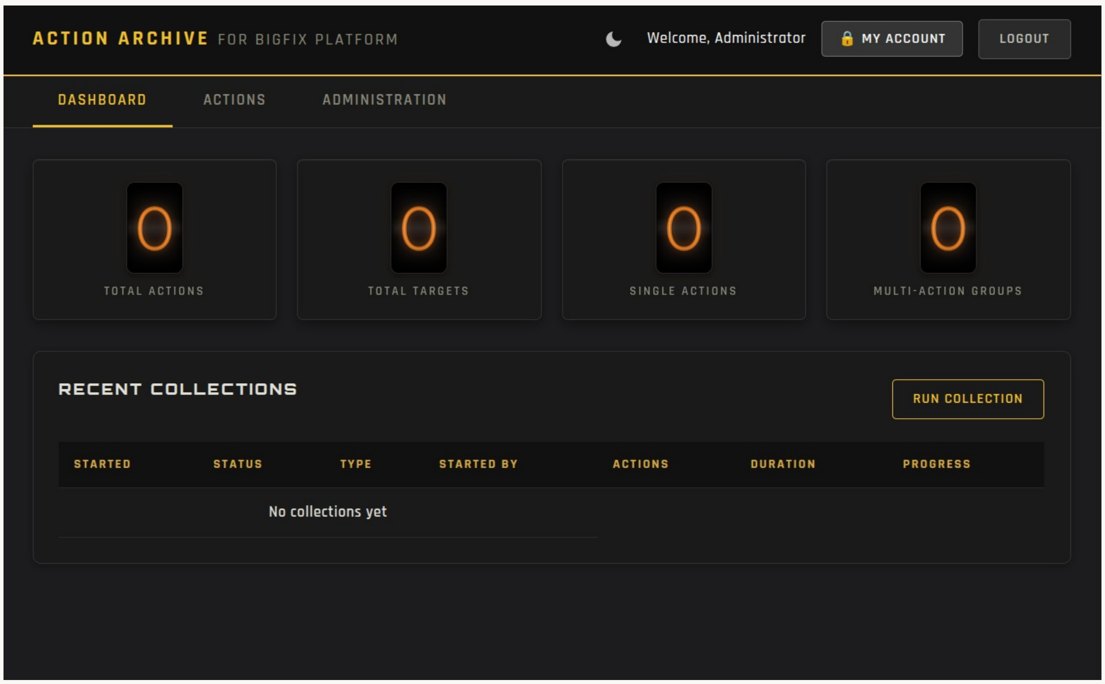

## 3.4 Default Credentials

> ❗ **IMPORTANT** Credentials are set during the installation process. There are no factory-default username or password values hard-coded in the application.

On first login, if the Must Change Password flag is set on your account, you will be prompted to choose a new password before reaching the dashboard.

## 3.5 First-Time Dashboard View

If no collection has been run, the dashboard will display zero counts across all stat cards and an empty Recent Collections table. To populate the archive, click Collect Now (see Section 4.3) or wait for a scheduled collection.



# 4. Dashboard

## 4.1 Stat Cards

The top of the dashboard displays four stat cards summarizing the current contents of the archive:

| Card | Description |
| --- | --- |
| Total Actions | Total count of all archived actions, including single actions and Multi-Action Groups. |
| Total Targets | Cumulative count of device-action assignments across all archived actions. |
| Single Actions | Count of single actions. |
| Multi-Action Groups | Count of parent MAG and Baseline records that contain component actions. |

## 4.2 Recent Collections Table

Below the stat cards, the Recent Collections table lists the last several collection runs:

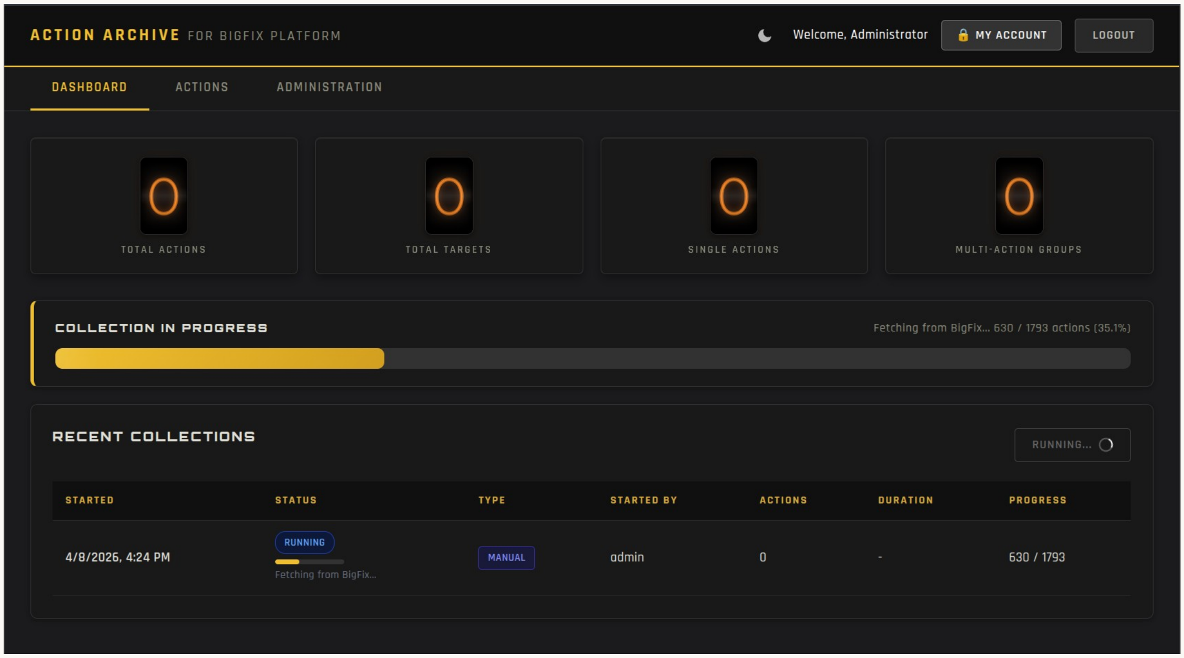
| Column | Description |
| --- | --- |
| Started | Date and time the collection began (in your local browser timezone). |
| Duration | Total elapsed time for the collection run. |
| Actions Found | Number of new actions discovered and archived in this run. |
| Status | Outcome: Completed, Running, or Error. |
| Details | Link to the full per-phase collection log for this run. |

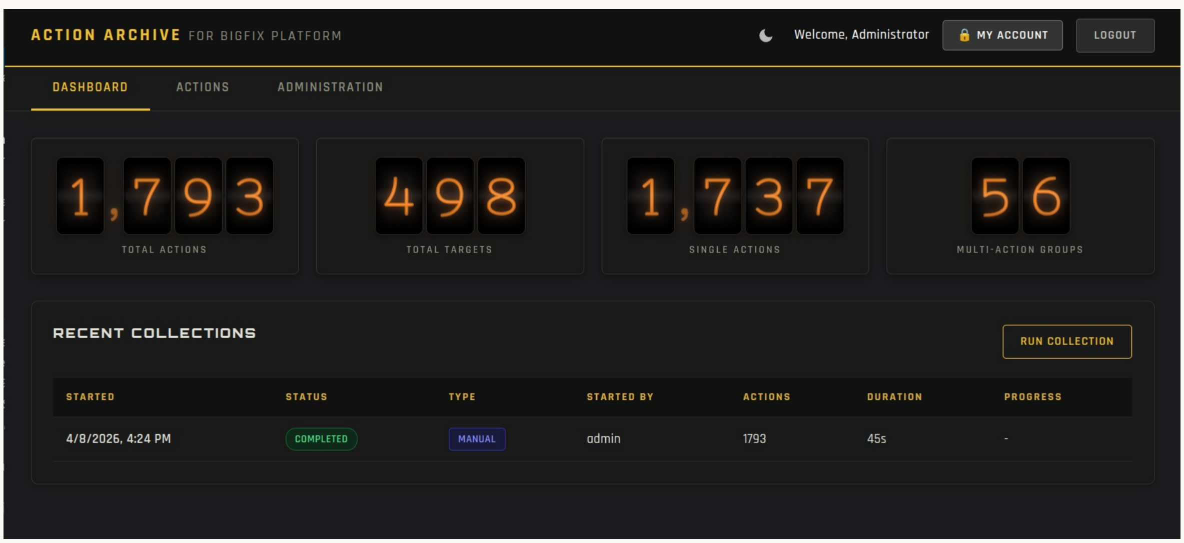
## 4.3 Run Collection Button

The Run Collection button triggers an immediate, on-demand collection and is available to Admin-role users only. During an active collection, the button is replaced by a live progress bar. If a scheduled collection is already running, the on-demand run is queued to start immediately after the current one completes.

## 4.4 Live Collection Progress

While a collection is in progress, a progress bar displays the current phase name (for example, Fetching or Storing) and a percentage completion. The bar disappears automatically on completion, and the stat cards reflect the updated counts.

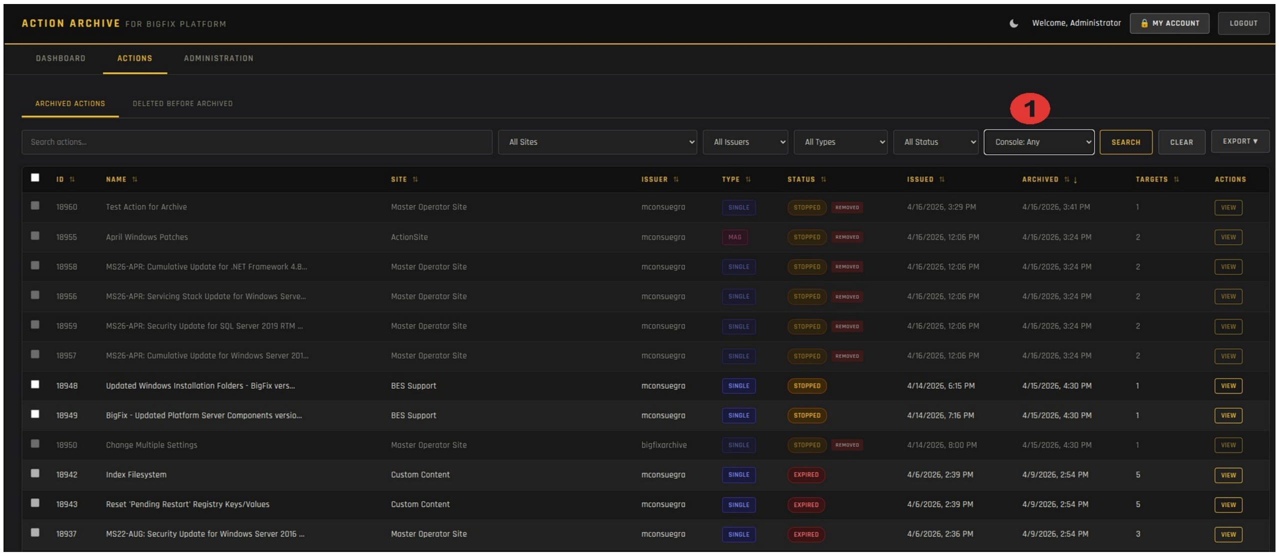

ACTION ARCHIVE FOR BIGFIX PLATFORM
Welcome, Administrator
MY ACCOUNT
LOGOUT
DASHBOARD
ACTIONS
ADMINISTRATION
0
TOTAL ACTIONS
0
TOTAL TARGETS
0
SINGLE ACTIONS
0
MULTI-ACTION GROUPS
COLLECTION IN PROGRESS
Fetching from BigFix... 630 / 1793 actions (35.1%)
RECENT COLLECTIONS
RUNNING...
STARTED
STATUS
TYPE
STARTED BY
ACTIONS
DURATION
PROGRESS
4/8/2026, 4:24 PM
RUNNING
Fetching from BigFix...
MANUAL
admin
0
-
630 / 1793


ACTION ARCHIVE FOR BIGFIX PLATFORM
Welcome, Administrator
MY ACCOUNT
LOGOUT
DASHBOARD
ACTIONS
ADMINISTRATION
1,793
TOTAL ACTIONS
498
TOTAL TARGETS
1,737
SINGLE ACTIONS
56
MULTI-ACTION GROUPS
RECENT COLLECTIONS
RUN COLLECTION
STARTED
STATUS
TYPE
STARTED BY
ACTIONS
DURATION
PROGRESS
4/8/2026, 4:24 PM
COMPLETED
MANUAL
admin
1793
45s
-

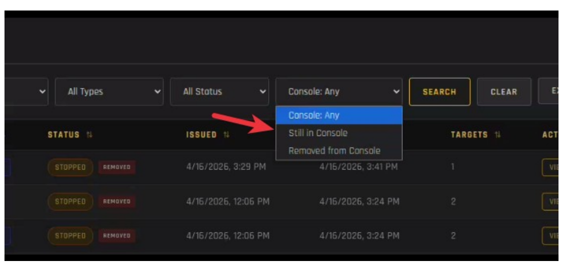
# 5. Browsing and Searching Actions

## 5.1 Actions Tab Overview

The Actions tab provides the primary interface for exploring archived actions. It opens to a paginated table listing all archived actions, with controls for searching, filtering, sorting, and exporting.




1 When actions are deleted from the console after they are collected they will display a "REMOVED" badge in the status field. This drop-down will allow filtering by All/Removed actions in the archive actions list.



## 5.2 Column Descriptions

| Column | Description | Notes |
|---|---|---|
| ID | BigFix-assigned numeric action identifier. | |
| Name | Display name as defined in BigFix. | |
| Type | Single Action or Multi-Action Group (MAG / Baseline). | |
| Site | Fixlet site the source fixlet belongs to. | May show N/A if source fixlet was deleted. |
| Issuer | BigFix operator who created the action. | |
| Status | Last known action status (Expired, Stopped, etc.). | |
| Issued | Date and time the action was created in BigFix. | Displayed in your local browser timezone. |
| Targets | Number of devices the action was targeted at. | |
| Archived | Date and time the action was collected into the archive. | |
| Actions | View button (click to see a detail of the action) | |

## 5.3 Searching and Filtering

The search bar accepts free-text queries and matches against Action ID, Action Name, Issuer, and Site simultaneously. Additional filter controls:

- Site: Restrict results to actions from a specific content site.
- Issuer: Filter to actions created by a specific BigFix operator.
- Type: Show only Single Actions or only Multi-Action Groups.
- Status: Filter by action status (Expired, Stopped, etc.).

All filters are additive. Click Reset Filters to clear all active selections.

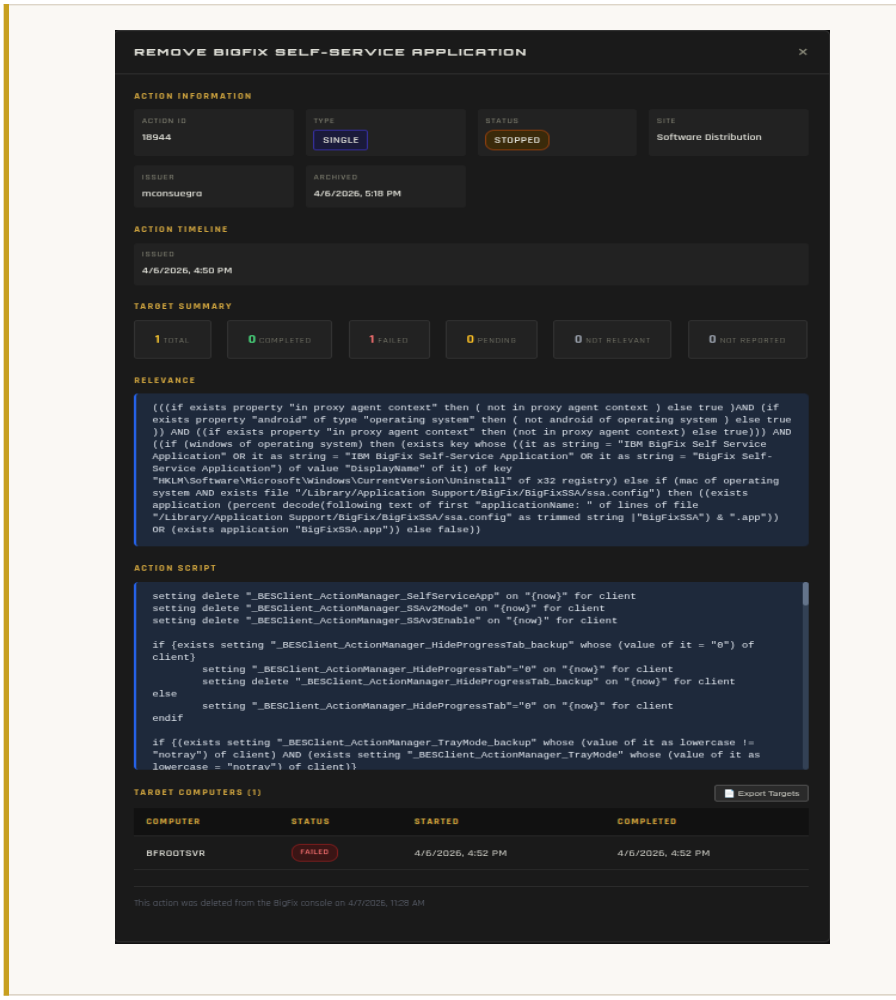
## 5.4 Sorting and Pagination

Click any column header to sort by that field; click again to reverse the sort order. The active sort is indicated by an arrow icon in the column header. Use the pagination controls at the bottom to navigate pages, and the page-size selector to change the number of rows displayed per page.

## 5.5 Action Detail View

Click the “View” button under the “Actions” column to open the detail view of the specific action. Action details include:

- Core properties: name, type, site, issuer, status, dates, Action Script and Relevance expression.
- Components: For Multi-Action Groups and Baselines, lists all member actions with individual IDs, names, and statuses. BigFix copies fixlet content inline when deploying baselines, so component records reflect content at deployment time.
- Targets: Every device the action was targeted at, with per-device status, last report time, and result message.



# 6. Deleted Actions Tracking

## 6.1 Overview

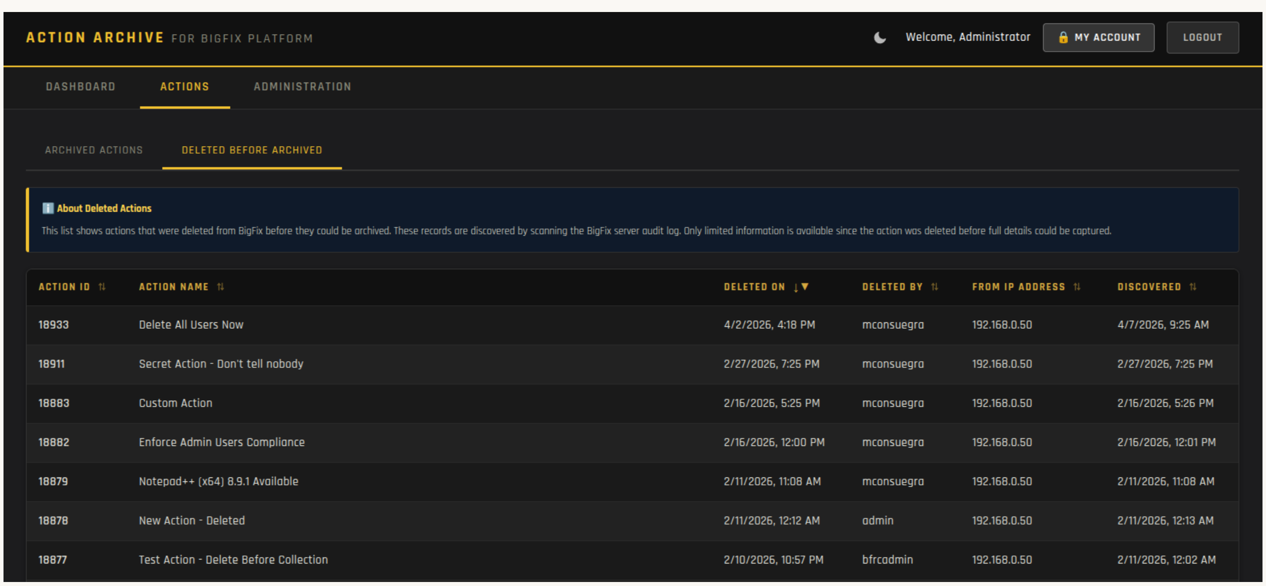
The Deleted Before Archived sub-tab within the Actions tab lists BigFix actions that were deleted from the BigFix server ***before Action Archive had the opportunity to collect them***. This provides visibility into a gap inherent to any periodic collection approach: actions created and deleted between collection cycles.

## 6.2 How Data Gets Here

The Audit Agent (see Section 11) monitors the BigFix server_audit.log file using a Windows Scheduled Task. When the agent detects an action deletion event, it submits that event to the Archive REST API using an API key. The record is stored even if the action's full details were never collected.

## 6.3 Fields Displayed

| Field | Description |
|---|---|
| Action ID | BigFix action identifier as recorded in the audit log. |
| Action Name | Action name from the audit log, if available. |
| Deleted On | Timestamp when the deletion was recorded in server_audit.log. |
| Deleted By | BigFix operator username that performed the deletion. |
| From IP Address | IP address of the BigFix Console from which the deletion was performed. |
| Discovered | Indicates when the action was archived. |



# 7. Exporting Data

Action Archive supports several export formats from different parts of the interface. All exports reflect the current filter state. Apply filters before exporting to scope the output.

| Export | Format | Contents | Accessed From |
|---|---|---|---|
| Action List | CSV | Filtered action metadata: ID, name, type, site, issuer, status, issued date, target count. | Actions tab → Export button |
| Action List | PDF | Formatted action summary with statistics header and full action table. | Actions tab → Export button |
| Target Computers | CSV | Per-device status for one action: device name, status, last report, result message. | Action detail → Targets tab |
| Audit Log | CSV | Full security event log: timestamp, user, event type, details, IP, user agent. | Administration → Logs → Audit Log |
| Audit Log | PDF | Formatted audit log with statistics header. | Administration → Logs → Audit Log |

> ℹ️ **NOTE** All exports honor the data-visibility scopes assigned to the current user. A Viewer-scoped user will only export records they are permitted to see.

# 8. Administration: User Management

User management is available from the Administration menu.

## 8.1 User Accounts

### Creating a User

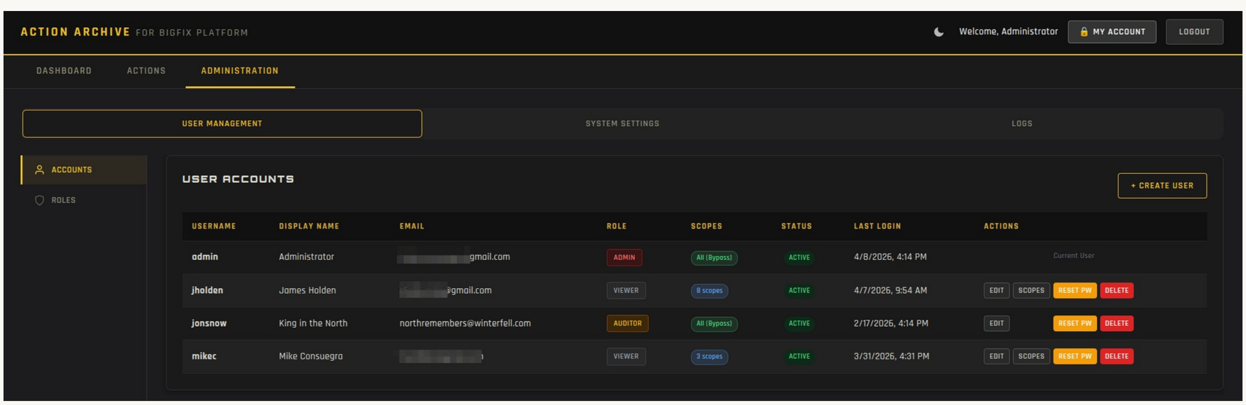
1. Navigate to Administration → Users.
2. Click Create User.
3. Complete all required fields: Username, Email Address, Display Name, Role, and Initial Password.
4. Optionally enable Must Change Password to force a password change on first login.
5. Click Save. The user can log in immediately.

### Editing a User

Click the Edit icon next to any user to modify their email address, display name, active status, or role. Username cannot be changed after creation. Changes take effect immediately.

### Deleting a User

Click the Delete icon and confirm. Deletion is permanent. Audit log entries associated with the user are retained with the username preserved in the record even after account deletion.




### Admin-Initiated Password Reset

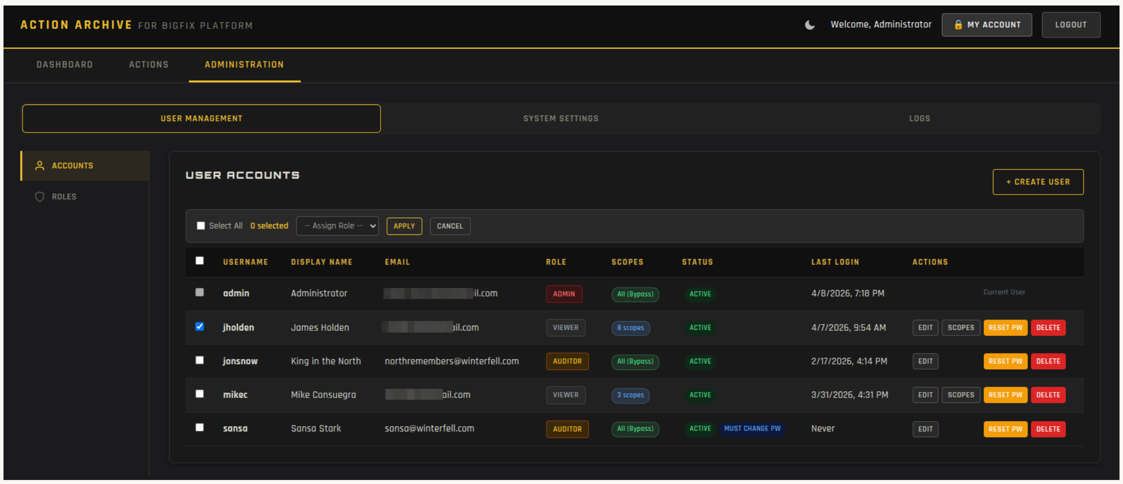
Click Reset Password next to a user account, enter a new temporary password, and save. The 'Must Change Password' flag is automatically set, forcing the user to choose a new password at their next login.

### Unlocking a Locked Account

Accounts are locked after the configured number of consecutive failed login attempts (see Section 9.2). Click Unlock next to the affected user to restore access immediately. Accounts also unlock automatically after the configured Lockout Duration.

### Bulk Role Assignment

Select multiple users using the checkboxes in the user list, then choose a new role from the Assign Role dropdown to apply it to all selected accounts at once.




## 8.2 Roles and Permissions

Action Archive uses three roles. The table below details each role:

| Capability | Admin | Auditor | Viewer |
|---|---|---|---|
| View archived actions and search | Yes | Yes | Yes |
| View action detail and target results | Yes | Yes | Yes |
| Export action lists (CSV / PDF) | Yes | Yes | Yes |
| View Deleted Before Archived | Yes | Yes | No |
| View and export Audit Log | Yes | Yes | No |
| Manage user accounts | Yes | No | No |
| Change system settings | Yes | No | No |
| Trigger Collect Now | Yes | No | No |
| Create and revoke API keys | Yes | No | No |
| Create and restore backups | Yes | No | No |

| Capability | Admin | Auditor | Viewer |
|---|---|---|---|
| View application and system logs | Yes | No | No |

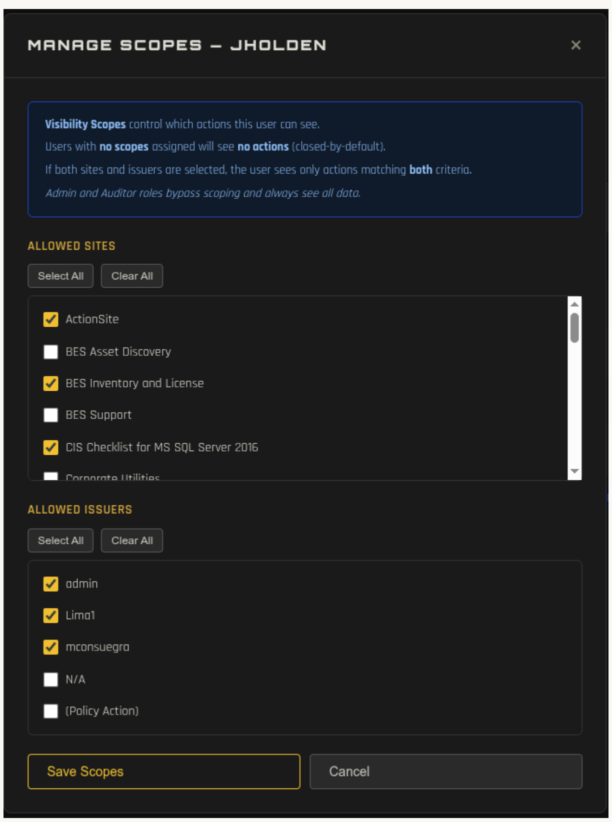
## 8.3 User Scopes (Data Visibility Restrictions)

Scopes allow administrators to restrict which actions a user can see based on Site or Issuer. A user with no scopes sees all archived actions. A user with scopes assigned sees only actions matching at least one of their scope conditions.

To assign scopes:

1. Open the user's record and click Manage Scopes.
2. Add one or more Site Name or Issuer conditions using the scope editor.
3. Click Save.

Scope enforcement applies to the action list, search results, action detail views, exports, and API calls. Example: members of the Patching team should see only actions from the "Patches for Windows" site. Assign a Site scope of that value to their accounts.



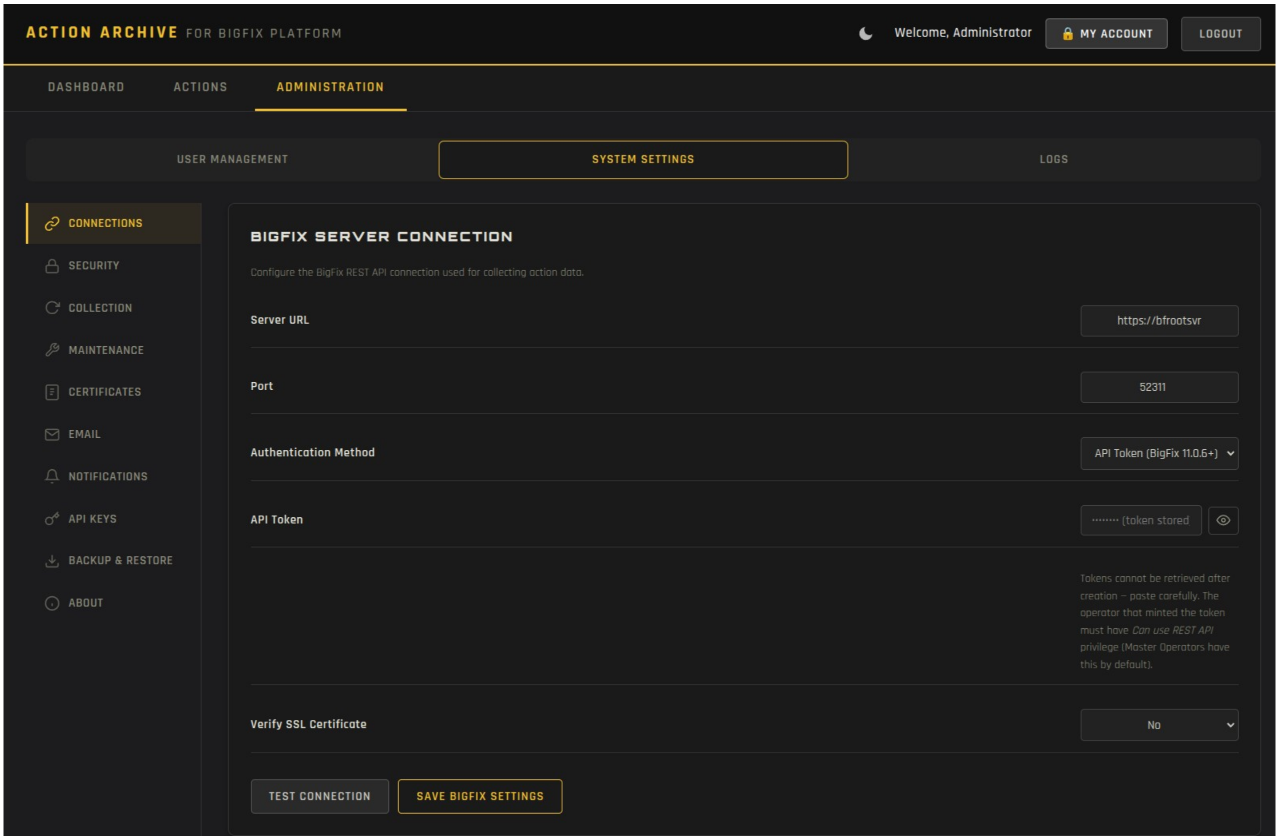
# 9. Administration: System Settings

All system settings are accessible to Admin-role users from Administration → Settings. Changes made through the web interface take effect immediately without a service restart unless otherwise noted.

## 9.1 Connections

| Setting | Description | Example / Default |
|---|---|---|
| BigFix Server URL | Hostname or IP of your BigFix Root Server. | 192.168.1.10 |
| BigFix Port | Port for the BigFix REST API. | 52311 |
| BigFix Username | BigFix Master Operator account used for API queries. | |
| BigFix Password | Password for the above account (encrypted at rest). | |
| Verify SSL Cert | Validate the BigFix server's SSL certificate. Disable for self-signed BigFix certs. | No (default) |
| BigFix Auth Method | Authentication method used to query the BigFix REST API. Password uses the Operator Username and Password (HTTP Basic). API Token uses a BigFix API token in an Authorization: Bearer header (recommended for non-MO operators and for installs that restrict interactive credentials). New in v1.1.0. | Password (default) |
| BigFix API Token | Required when Auth Method is API Token. Paste the token once. it is encrypted at rest and cannot be retrieved through the UI. To mint a token, use the BigFix Console: Tools → Manage Tokens. The underlying operator must retain the REST API (Can use REST API) permission. | (encrypted at rest) |
| DB Host | PostgreSQL server hostname. | localhost |
| DB Port | PostgreSQL listen port. | 5432 |
| DB Name | Name of the archive database. | bigfix_archive |
| DB Username | PostgreSQL user used by the application. | bigfix_user |
| DB Password | Database password (encrypted at rest). | |

> ❗ **IMPORTANT** The operator account used must have Master Operator (MO) privileges. Without MO access, Session Relevance queries used to discover actions across all sites will return incomplete results.




When Auth Method is Password, the BigFix Username and Password fields apply; when Auth Method is API Token, only the BigFix API Token field applies. Changing auth methods at any time is safe. Credentials for the unused method are preserved encrypted.

## 9.2 Security Settings

### Password Policy

| Setting | Description | Default |
|---|---|---|
| Minimum Length | Minimum number of characters required. | 8 |
| Require Uppercase | At least one uppercase letter must be present. | Yes |
| Require Lowercase | At least one lowercase letter must be present. | Yes |
| Require Numbers | At least one digit must be present. | Yes |
| Require Special Chars | At least one special character must be present. | Yes |

| Setting | Description | Default |
| --- | --- | --- |
| Password Expiration | Days after which passwords must be changed. 0 = never. | 0 |
| Password History Count | Prevents reuse of the last N passwords. 0 = no history. | 0 |

## Account Lockout Policy

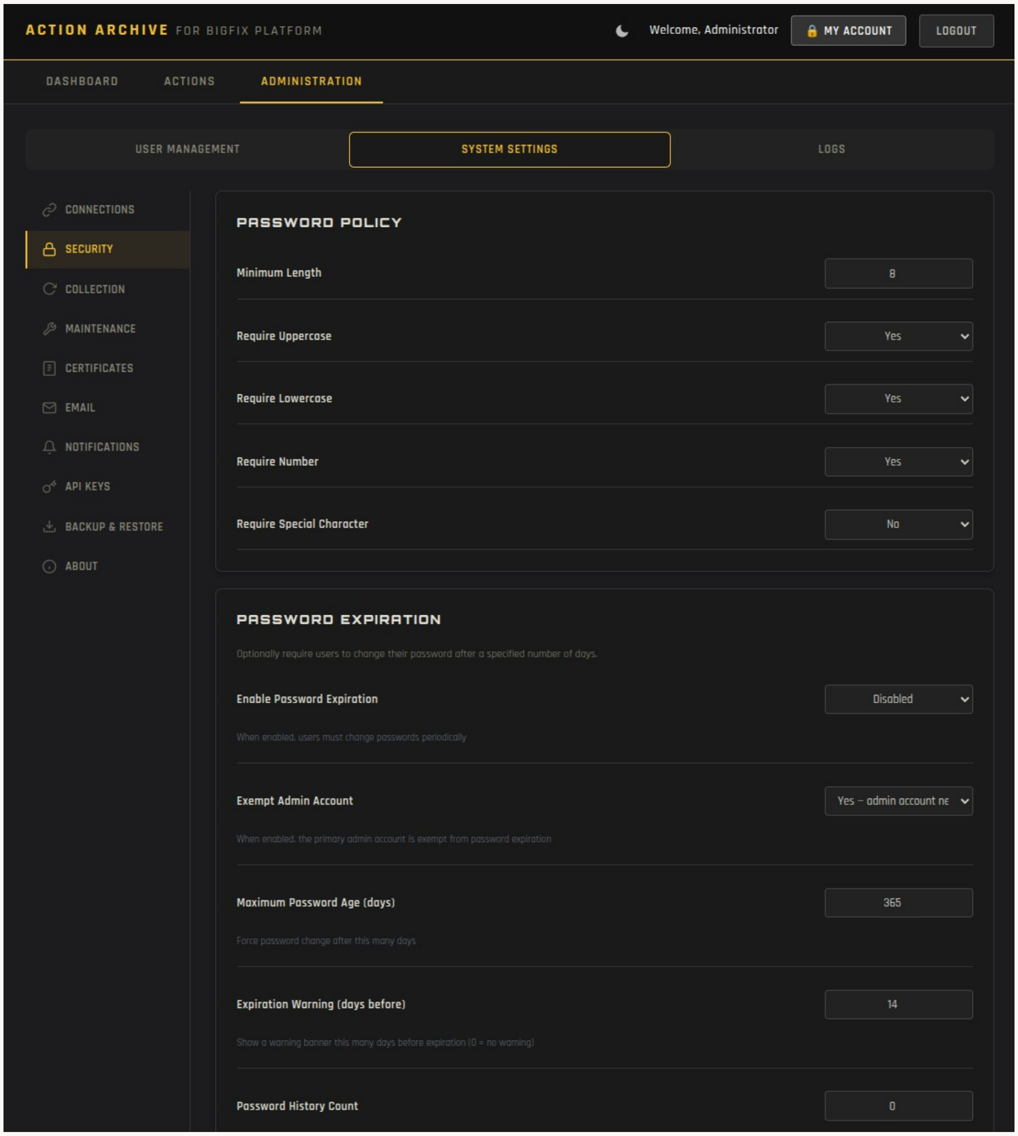
| Setting | Description | Default |
| --- | --- | --- |
| Max Failed Attempts | Consecutive failed logins before account is locked. | 5 |
| Lockout Duration | Minutes the account remains locked. Admin can unlock manually. | 15 |

## Rate Limiting

| Setting | Description | Default |
| --- | --- | --- |
| Max Attempts by IP | Number of failed login attempts from a specific IP before it is blocked from attempting to logon. | 10 |
| Time Window | Failed attempts during this time window count towards rate limit | 15 |
| Ban duration | IP is banned from re-attempting for this time. | 15 |
| IP Whitelist | List of IPS not subject to rate limiting | |

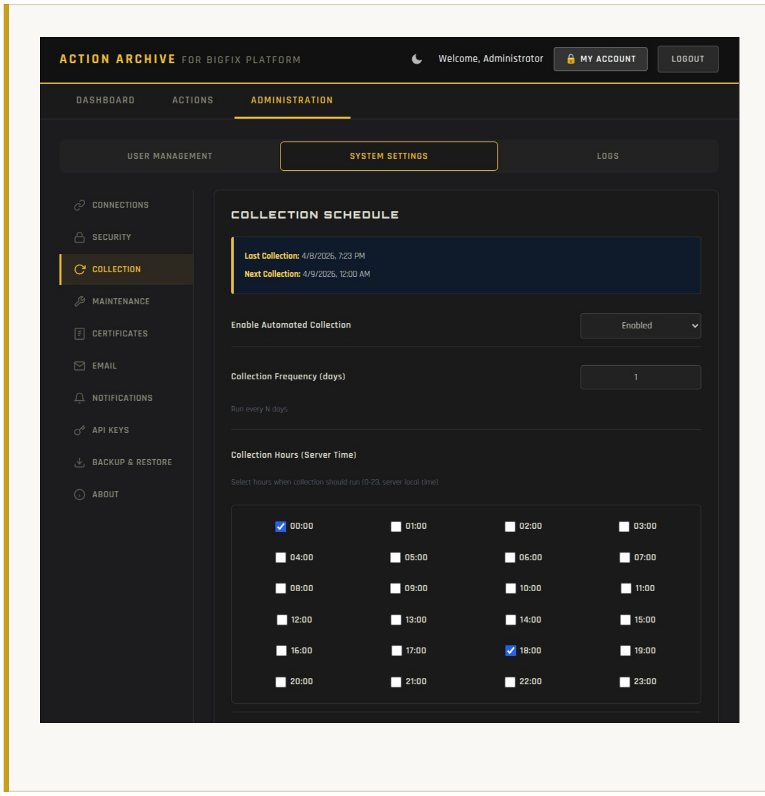
## Login Page Customization

| Setting | Description | Default |
| --- | --- | --- |
| Banner Message | Message that will display on the logon page if feature is enabled | |
| Banner Style | Info / Warning / Legal styles | |
| Consent Text | Message presented to user for consideration before logging in. | |
| Require Consent | If enabled, user must acknowledge acceptance of consent message | |

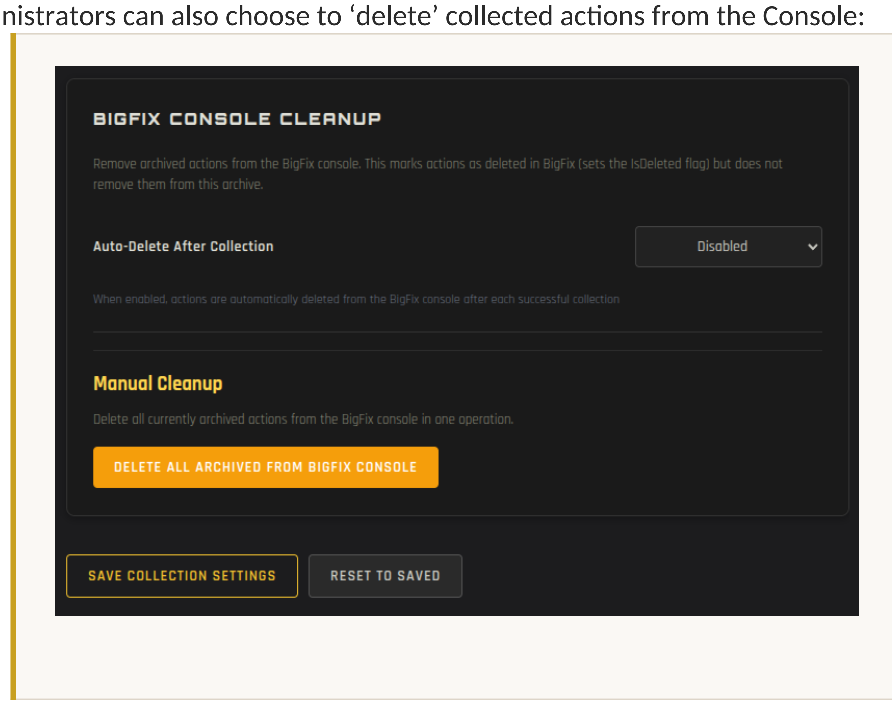

ACTION ARCHIVE FOR BIGFIX PLATFORM
Welcome, Administrator
MY ACCOUNT
LOGOUT
DASHBOARD
ACTIONS
ADMINISTRATION
USER MANAGEMENT
SYSTEM SETTINGS
LOGS
CONNECTIONS
SECURITY
COLLECTION
MAINTENANCE
CERTIFICATES
EMAIL
NOTIFICATIONS
API KEYS
BACKUP & RESTORE
ABOUT
PASSWORD POLICY
Minimum Length
8
Require Uppercase
Yes
Require Lowercase
Yes
Require Number
Yes
Require Special Character
No
PASSWORD EXPIRATION
Optionally require users to change their password after a specified number of days.
Enable Password Expiration
Disabled
When enabled, users must change passwords periodically
Exempt Admin Account
Yes – admin account ne
When enabled, the primary admin account is exempt from password expiration
Maximum Password Age (days)
365
Force password change after this many days
Expiration Warning (days before)
14
Show a warning banner this many days before expiration (0 = no warning)
Password History Count
0

## 9.3 Collection Schedule

- Enable Automatic Collection: Master toggle to enable or disable scheduled collections.
- Frequency (Every N Days): Run a collection every N days from the last completed run.
- Collection Hours: Specify the hour(s) of the day in which collection will occur.



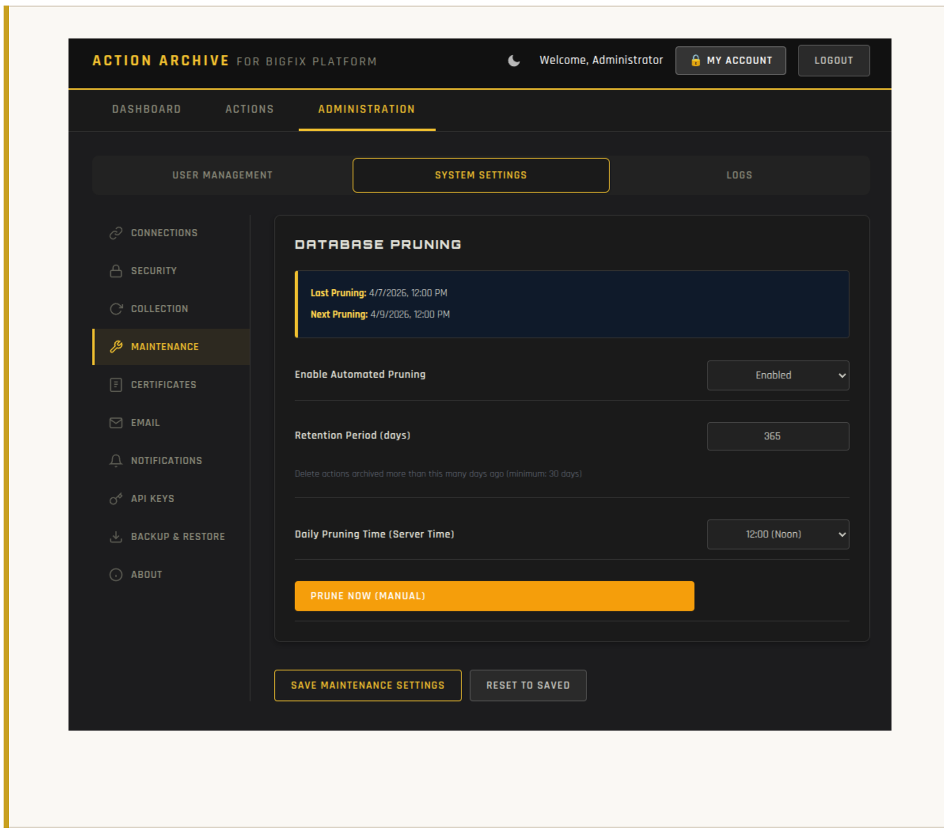

Administrators can also choose to 'delete' collected actions from the Console:



## 9.4 Maintenance (Database Pruning)

- Retention Period (Days): Actions archived more than this many days ago are removed by the pruning job.
- Run Pruning Now: Triggers an immediate pruning operation using the current retention setting.
- Pruning History: A table showing past runs with timestamps and row counts deleted.

> ⚠️ **WARNING** Pruning permanently deletes action records from the archive database. This operation cannot be undone. Ensure your retention period aligns with your organization's data retention policies. Create a database backup (Section 9.9) before running pruning for the first time.



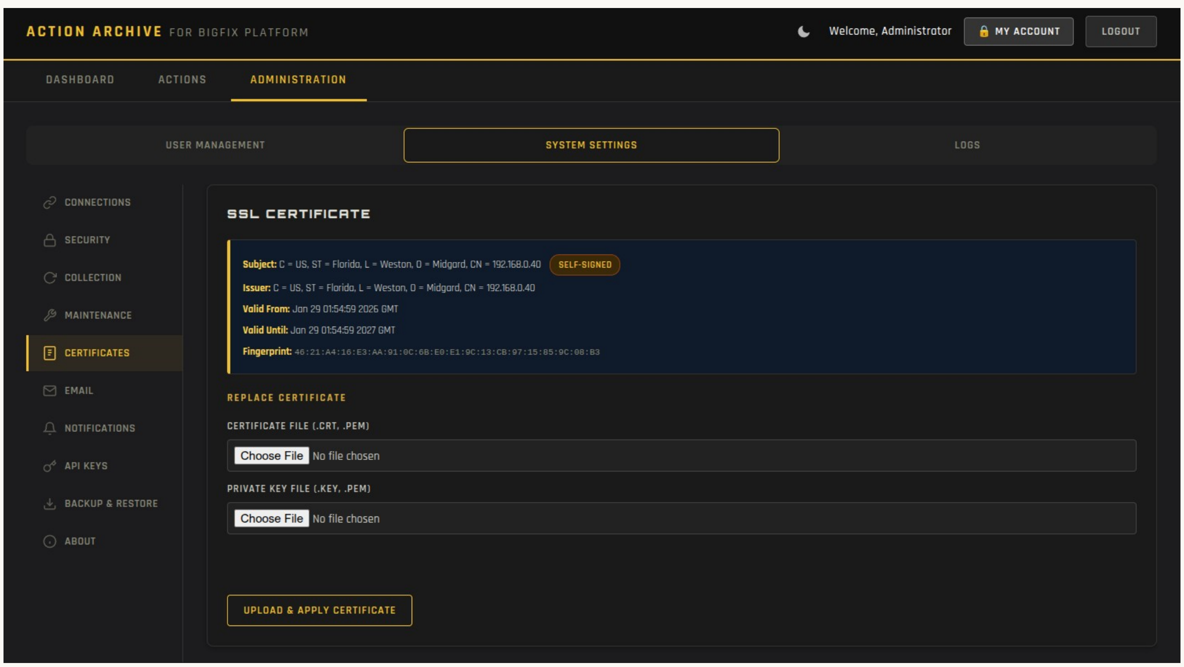
## 9.5 Certificates (SSL)

This panel displays information about the current SSL certificate used by Nginx and allows you to replace it through the web interface.

- Current Certificate Details: Displays issuer, subject, validity dates, and SHA-256 fingerprint of the active certificate.
- Replace Certificate: Upload a PEM-encoded X.509 certificate chain and corresponding private key. Nginx is reloaded automatically after upload.

> ℹ️ **NOTE** This panel manages the Archive Server's own SSL certificate (what browsers see). It does not affect the BigFix server's certificate. To control BigFix certificate validation, use the Verify SSL toggle in Connections (Section 9.1).

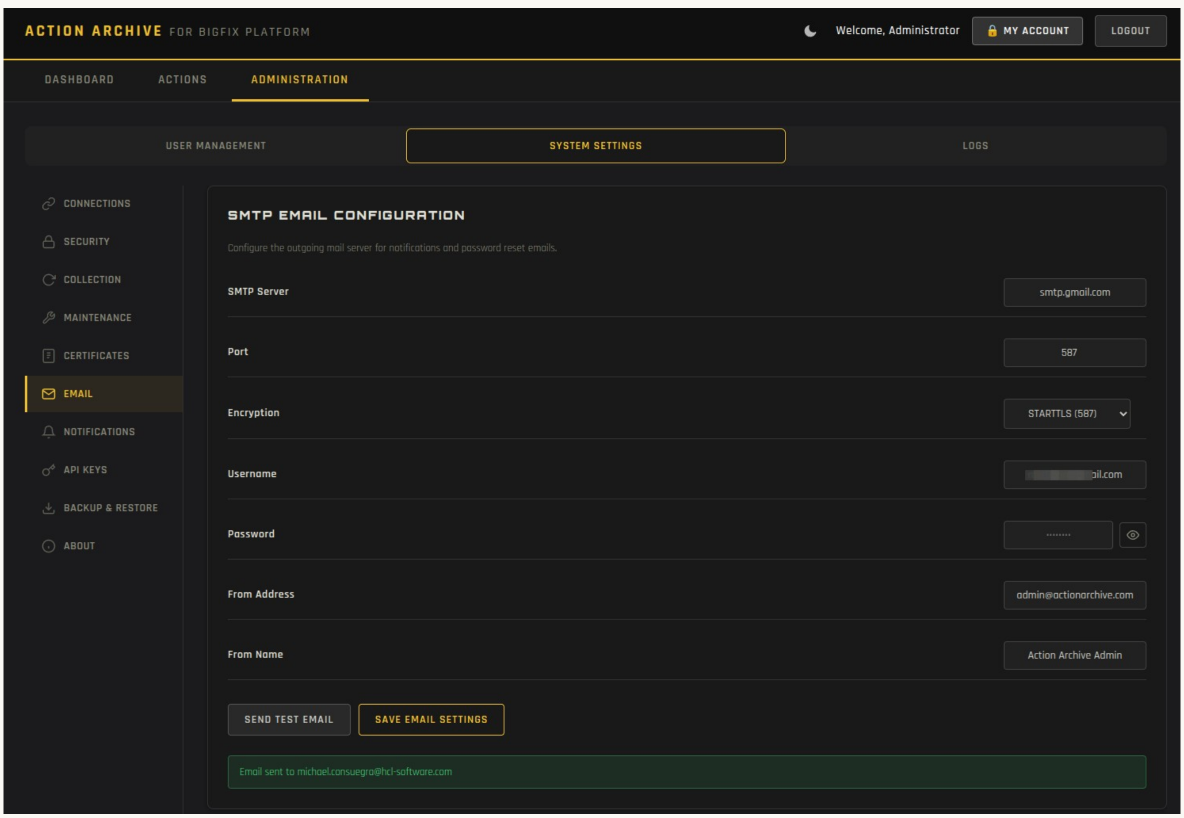
> ❗ **IMPORTANT** The certificate and private key must be in PEM format. The certificate file should include the full chain (server cert + any intermediate CAs), ordered from leaf to root.



## 9.6 Email / SMTP

| Field | Description | Example |
| --- | --- | --- |
| SMTP Server | Hostname of your mail relay server. | smtp.example.com |
| Port | Port for the SMTP connection. | 587 |
| Username | SMTP authentication username. | archive@example.com |
| Password | SMTP password (encrypted at rest). | |
| Encryption | Connection security: none, starttls, or ssl. | starttls |
| From Address | Sender address shown on outgoing messages. | archive@example.com |
| From Name | Sender display name shown in email clients. | Action Archive |

Use the Send Test Email button to verify your SMTP configuration before saving.

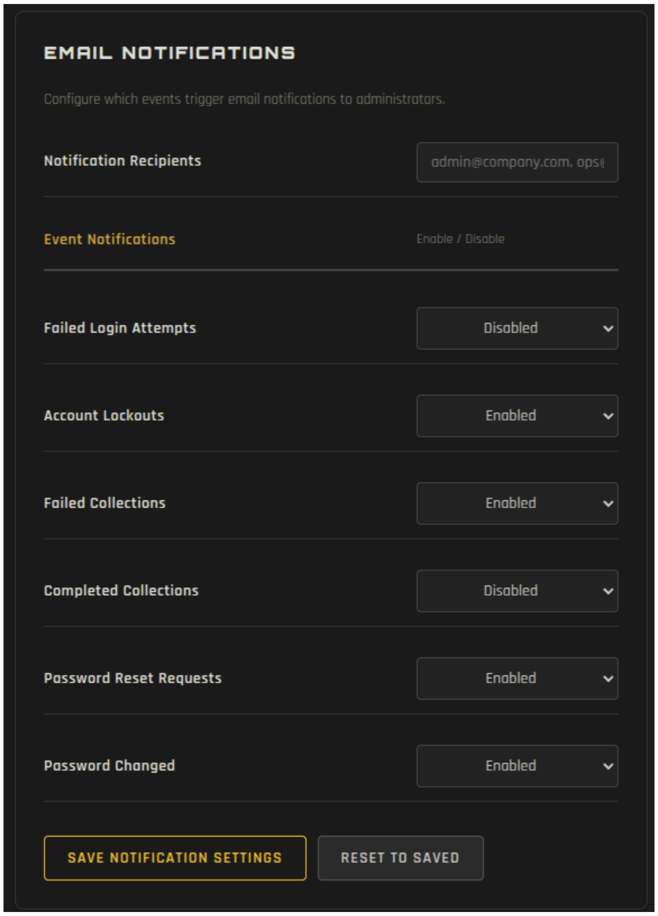


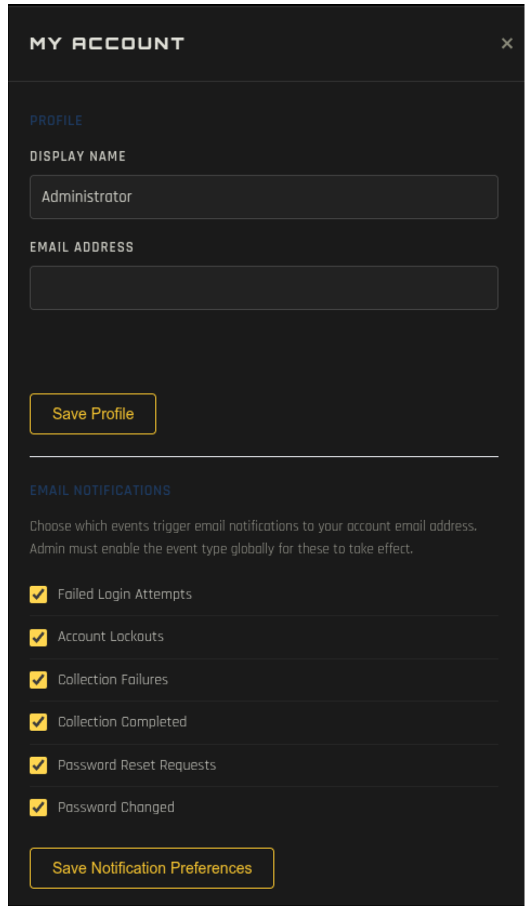
## 9.7 Notifications

| Event Type | Description |
|---|---|
| Completed Collection | Sent after each successful collection run with a summary of actions archived. |
| Failed Collections | Sent when a collection run encounters an unrecoverable error. |
| Account Lockouts | Sent when a user account is locked out (too many invalid login attempts) |
| Password Reset Requests | Confirmation sent to the user when their password is reset. |
| Password Changed | Confirmation sent to the user when their password is reset. |
| Failed Login Attempts | Alerts on repeated failed login attempts or account lockouts. |

Each notification email is sent only to the target user . A Failed Login Attempt alert goes to the account that was targeted, an Account Lockouts notice goes to the locked-out user, and a Completed Collection or Failed Collections email goes to the user who triggered that collection run. For scheduled or system-initiated events where no human initiator exists, the Additional Recipients list (a comma-separated set of fallback addresses) is used instead. Users manage their per-event preferences from the My Account page. An email is delivered only when the matching master switch above is enabled, the targeted user has the per-event preference turned on, and an email address is on file for that user.

The Password Reset Requests switch additionally controls the self-service Forgot Password flow. When it is disabled, a user who requests a reset still sees the same confirmation message, but no reset email is sent and the request is recorded in the audit log. In that configuration, self-service reset is effectively turned off and an administrator resets passwords from the User Management page. The switch is enabled by default.




## 9.8 API Keys

API keys provide machine-to-machine authentication for the Archive REST API without interactive login. The Audit Agent uses an API key to submit discovered deleted actions.

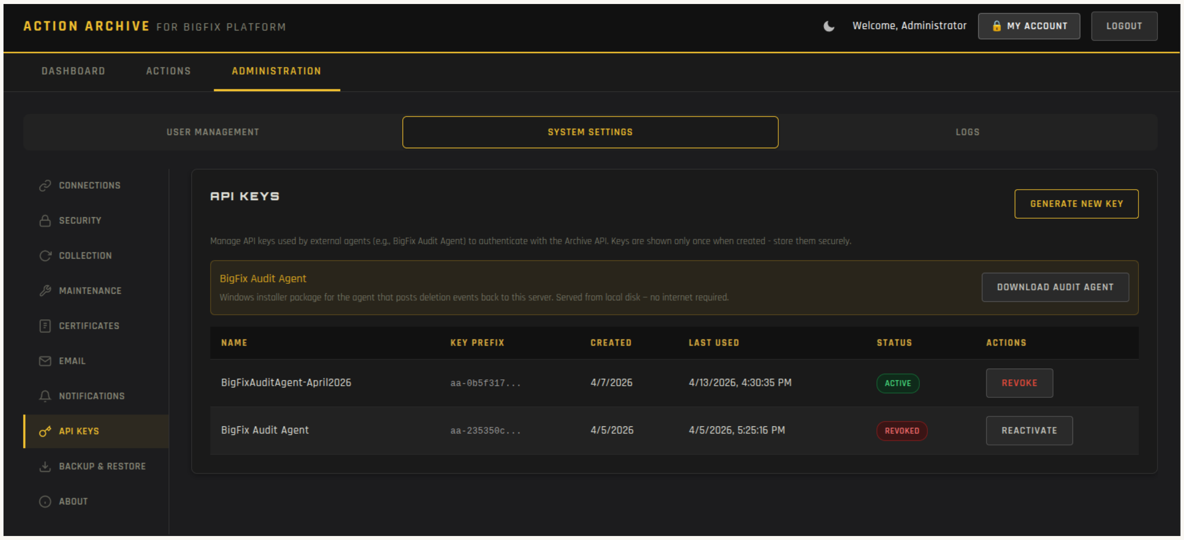
To generate a new API key:

1. Navigate to Administration → Settings → API Keys.
2. Click Generate New Key.
3. Enter a descriptive name (e.g., Audit Agent for BigFix Server 01).
4. Click Generate.
5. Copy the full key from the confirmation dialog immediately.

> ❗ **IMPORTANT** The full API key is displayed only once at creation. The system stores only the key prefix for reference. If the key is lost, revoke it and generate a new one. Store API keys in a secrets manager or protected configuration file.

API requests authenticate by including the key in the X-API-Key HTTP header. Keys can be individually revoked (disabled). Revoked keys remain visible in the list for audit purposes.

The API Keys panel also exposes a Download Audit Agent button. Clicking it streams the bundled Windows installer zip (action-archive-audit-agent-latest.zip) directly from the Archive Server's local disk. No outbound internet access is required. A download event is recorded in the Audit Log as AUDIT_AGENT_DOWNLOADED. See the Installation Guide, Part 3, for how to install the agent on the BigFix Root Server.




## 9.9 Backup, Restore, and Migration

### Creating a Backup

1. Click Create Backup Now.
2. The application runs pg_dump against the archive database.
3. The new backup file appears in the list with a timestamp and file size when complete.
4. Click the download icon to save the file to your local machine.

### Restoring from a Backup

1. Click Restore next to the desired backup file, or upload a previously downloaded file.
2. Review the confirmation prompt carefully.
3. Click Confirm Restore.

> ⚠️ **WARNING** Restoring a backup replaces ALL current database contents, including user accounts, system settings, archived actions, and audit logs. This operation cannot be undone. Always create a current backup before performing a restore.

Old backup files can be deleted from the interface to free disk space. Files are stored at /opt/bigfix-archive/backups/ on the server.

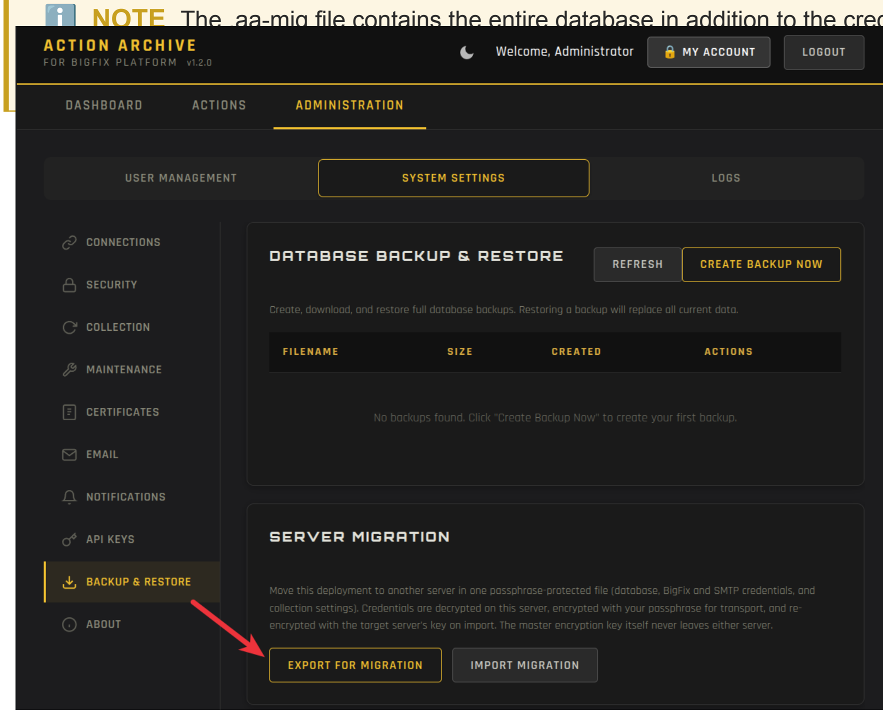
Server Migration is a related workflow used when moving an Action Archive deployment to a different host, such as replacing the underlying server hardware or moving from a 22.04 host to a fresh 24.04 or 26.04 install. Unlike the backup and restore flow above, it also carries the encrypted BigFix and SMTP credentials across hosts. The credentials are decrypted on the source server with that machine's key, packaged inside a passphrase-encrypted file (.aa-mig), and re-encrypted on the target server with the new machine's key. The passphrase is supplied by the operator at export time and is never stored.

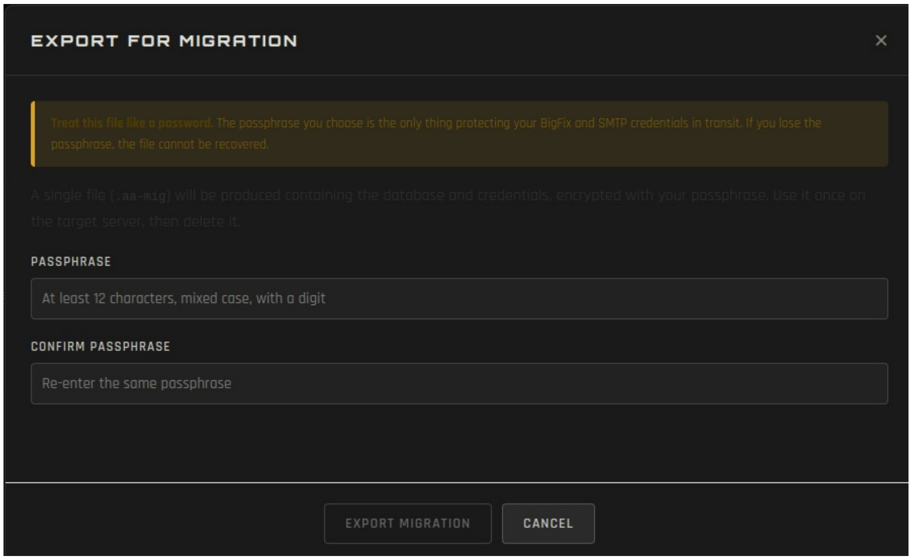
[SCREENSHOT PLACEHOLDER: The Server Migration panel inside Administration > Backup, showing the Export Migration Bundle and Import Migration Bundle buttons. Replace with actual screenshot before PDF export.]

## Exporting a Migration Bundle

1. Click Export Migration Bundle in the Server Migration panel of the Backup tab.
2. Enter a passphrase and confirm it. The passphrase must be at least 12 characters and include at least one uppercase letter, one lowercase letter, and one digit.
3. Click Export. The system runs pg_dump, decrypts the configured credentials with the source machine's key, encrypts the resulting envelope with the passphrase, and downloads the file as action-archive-migration-<timestamp>.aa-mig.
4. Record the passphrase before transferring the bundle. The passphrase is not stored on the source server and is not embedded in the bundle. If it is lost, the bundle cannot be opened.

ℹ️ **NOTE** The .aa-mig file contains the entire database in addition to the credentials. Treat it with the same ... e local copy from the source




EXPORT FOR MIGRATION
×
Treat this file like a password. The passphrase you choose is the only thing protecting your BigFix and SMTP credentials in transit. If you lose the passphrase, the file cannot be recovered.
A single file (.aa-mig) will be produced containing the database and credentials, encrypted with your passphrase. Use it once on the target server, then delete it.
PASSPHRASE
At least 12 characters, mixed case, with a digit
CONFIRM PASSPHRASE
Re-enter the same passphrase
EXPORT MIGRATION
CANCEL

## Importing a Migration Bundle

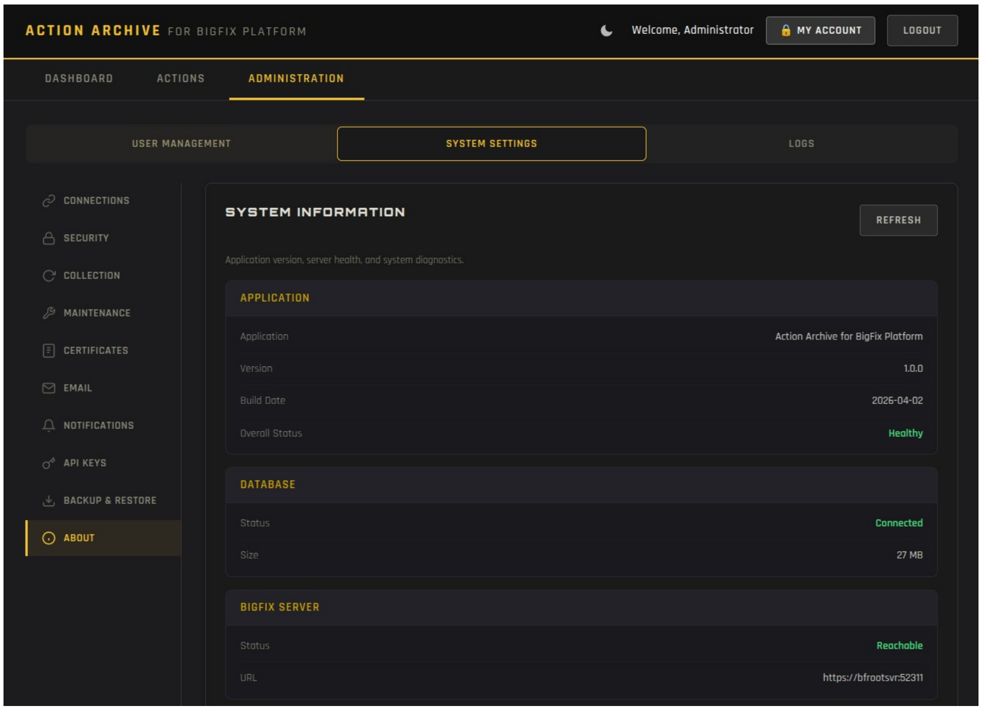
1. On the target server, complete a fresh install of Action Archive at the same version as the source, or a newer version.
2. Log in as administrator and navigate to Administration > Backup.
3. Click Import Migration Bundle in the Server Migration panel.
4. Select the .aa-mig file, enter the passphrase, and click Import.
5. The system verifies the envelope, decrypts the contents, terminates open connections to the application database, replaces the database with the imported contents, re-encrypts the credentials with the target machine's key, and updates the configuration file.

> ⚠️ **WARNING** Import is destructive. The existing application database on the target server is dropped and replaced with the contents of the bundle. Any data collected on the target between the fresh install and the import is lost. Run import only on a server you intend to overwrite.

> ❗ **IMPORTANT** If the passphrase is incorrect or the bundle has been tampered with, the system rejects the file without modifying the database. The database is only replaced after the bundle has been decrypted and verified.

## 9.10 About

The About panel displays application metadata and live system health information: version and build date, server uptime, disk usage, memory usage, database size, and integrity verification status (see Section 12.5).

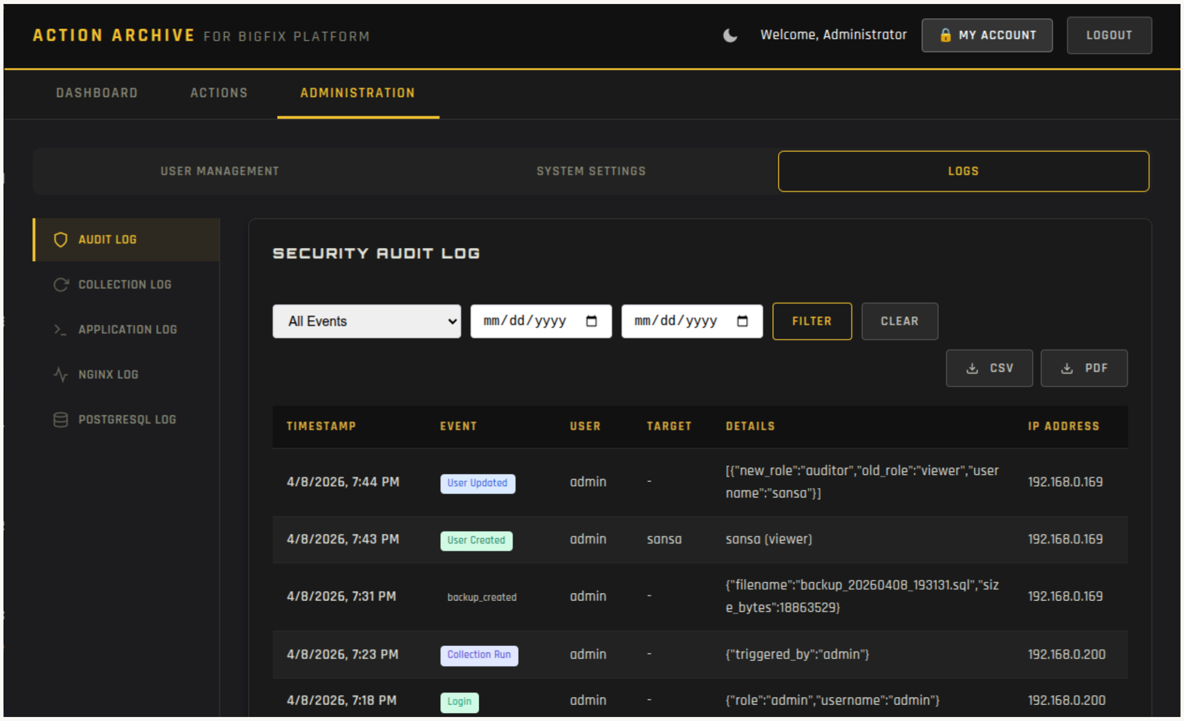


# 10. Administration: Logs

## 10.1 Audit Log

The Audit Log records every security-relevant event in the application and is available to Admin and Auditor roles from Administration → Logs → Audit Log.

| Event Type | What Triggers It |
|---|---|
| LOGIN_SUCCESS | Successful user authentication. |
| LOGIN_FAILURE | Failed login attempt (wrong password or unknown user). |
| LOGOUT | User-initiated or session-expiry logout. |
| ACCOUNT_LOCKED | Account locked after exceeding max failed attempts. |
| ACCOUNT_UNLOCKED | Admin manually unlocked a locked account. |
| USER_CREATED | New user account created. |
| USER_UPDATED | User account details modified. |
| USER_DELETED | User account permanently deleted. |
| PASSWORD_RESET | Admin-initiated or self-service password reset. |
| SETTINGS_CHANGED | Any system setting modified. |
| COLLECTION_TRIGGERED | Manual or scheduled collection started. |
| COLLECTION_COMPLETED | Collection run finished successfully. |
| COLLECTION_FAILED | Collection run terminated with an error. |
| BACKUP_CREATED | Database backup file created. |
| BACKUP_RESTORED | Database backup restored. |
| PRUNING_RUN | Data pruning operation executed. |
| EXPORT_GENERATED | User generated a CSV or PDF export. |
| API_KEY_CREATED | New API key generated. |
| API_KEY_REVOKED | API key disabled or deleted. |
| CERTIFICATE_REPLACED | SSL certificate updated via the web UI. |
| AUDIT_AGENT_DOWNLOADED | An administrator downloaded the bundled Audit Agent installer from the API Keys panel. |

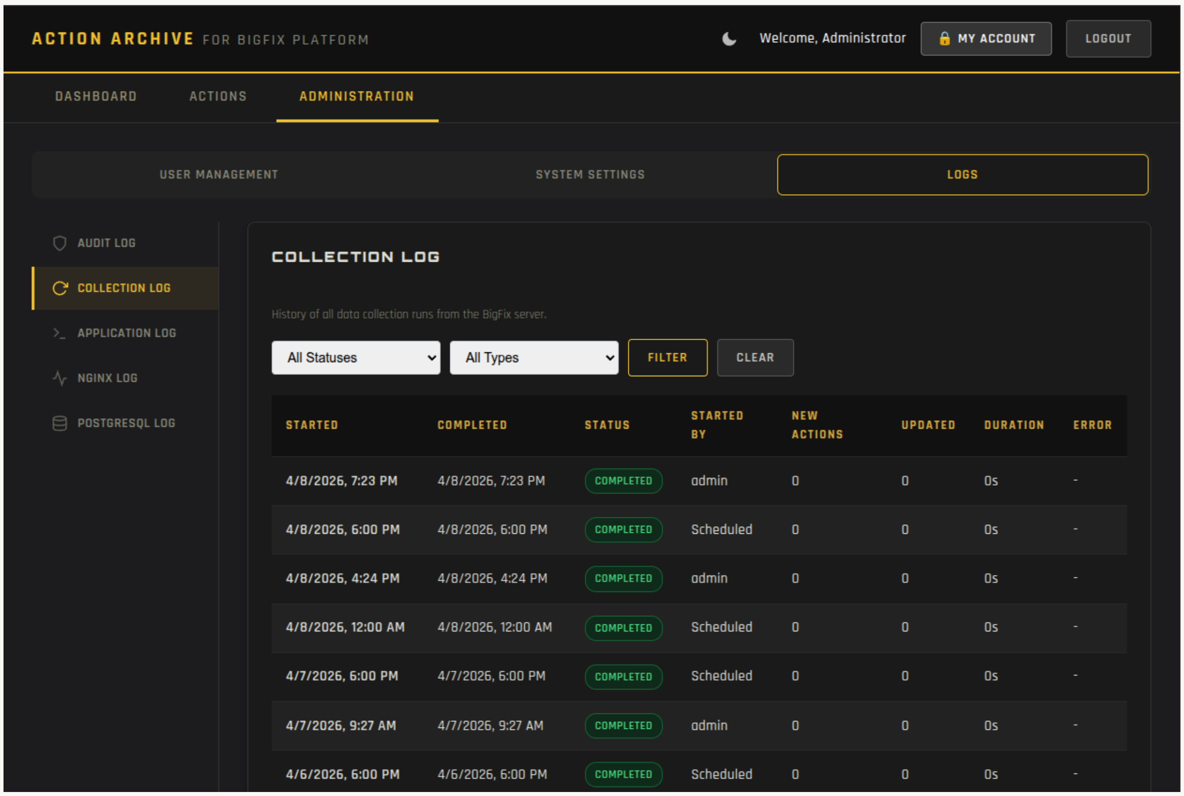



## 10.2 Collection Logs

Each collection run produces a detailed per-phase log accessed via the Details link in the Recent Collections table. The following phases are recorded:

| Phase | Description |
|---|---|
| Initializing | Establishing connection to the BigFix REST API and Session Relevance endpoint. Verifying credentials. |
| Discovery | Running Session Relevance query to enumerate all expired and stopped actions across all sites. |
| Filtering | Comparing discovered action IDs against the database to identify new actions not yet archived. |
| Metadata | Bulk-fetching supplemental data: site names, operator display names, and parent MAG IDs. |
| Fetching | Retrieving full action XML and target-device results for each new action via the REST API. |
| Storing | Inserting or upserting action records, components, and target results into PostgreSQL. |
| Complete | Collection finished. Summary of actions archived and elapsed time logged. |

| Status | Meaning |
|---|---|
| Completed | The collection ran without errors and all discovered actions were stored. |
| Running | A collection is currently in progress. |
| Error | The collection failed. Review the per-phase log for the specific error. |




## 10.3 Application, Nginx, and PostgreSQL Logs

- Application Log: Output from the FastAPI / Uvicorn process. Diagnose application errors, API exceptions, or startup failures. Equivalent to: journalctl -u bigfix-archive
- Nginx Access Log: All HTTP requests processed by Nginx, including response codes and sizes. Path: /var/log/nginx/bigfix-archive-access.log
- Nginx Error Log: Nginx-level errors, including upstream connection failures. Path: /var/log/nginx/bigfix-archive-error.log
- PostgreSQL Log: Database-level events, query errors, and connection issues.

# 11. Audit Agent Administration

## 11.1 Overview

The Action Archive Audit Agent is a lightweight Python utility that runs on the BigFix Root Server (Windows) as a Windows Scheduled Task. It monitors the BigFix server_audit.log file for action deletion events and reports them to the Archive REST API, preserving a record of actions deleted before they could be archived.

## 11.2 Configuration

The agent is configured via config.ini in its installation directory. Edits take effect on the next scheduled run. No restart is required.

As of v1.1.0, the agent installer ships inside the Archive Server tarball and is downloaded from the Archive web UI at Administration → System Settings → API Keys → Download Audit Agent. The installer writes config.ini into the agent install directory on the BigFix Root Server; subsequent configuration changes are made by editing that file directly.

| Setting | Description | Example / Default |
|---|---|---|
| api_url | Full HTTPS URL of the Archive Server. | https://192.168.0.40 |
| api_key | API key generated in Administration → Settings → API Keys. | aa-xxxxxxxxxxxx |
| audit_log_path | Full path to the BigFix server_audit.log file. | Auto-detected |
| position_file | File tracking the agent's last-read position in the audit log. | position.json |
| log_file | Path to the agent's own operational log. | agent.log |
| log_level | Logging verbosity: DEBUG, INFO, WARNING, or ERROR. | INFO |

Log level guidance: DEBUG produces verbose output for troubleshooting; INFO (default) records each successful submission; WARNING captures non-fatal issues; ERROR records failures requiring attention.

## 11.3 Managing the Scheduled Task

1. Open Task Scheduler on the BigFix Root Server.
2. Locate Action Archive Audit Agent in the Task Scheduler Library.
3. Right-click to Run (manual trigger), Disable, Enable, or modify the trigger interval.

To uninstall, run the installer batch file and choose the Uninstall option. This removes the scheduled task but does not delete the installation directory or log files.

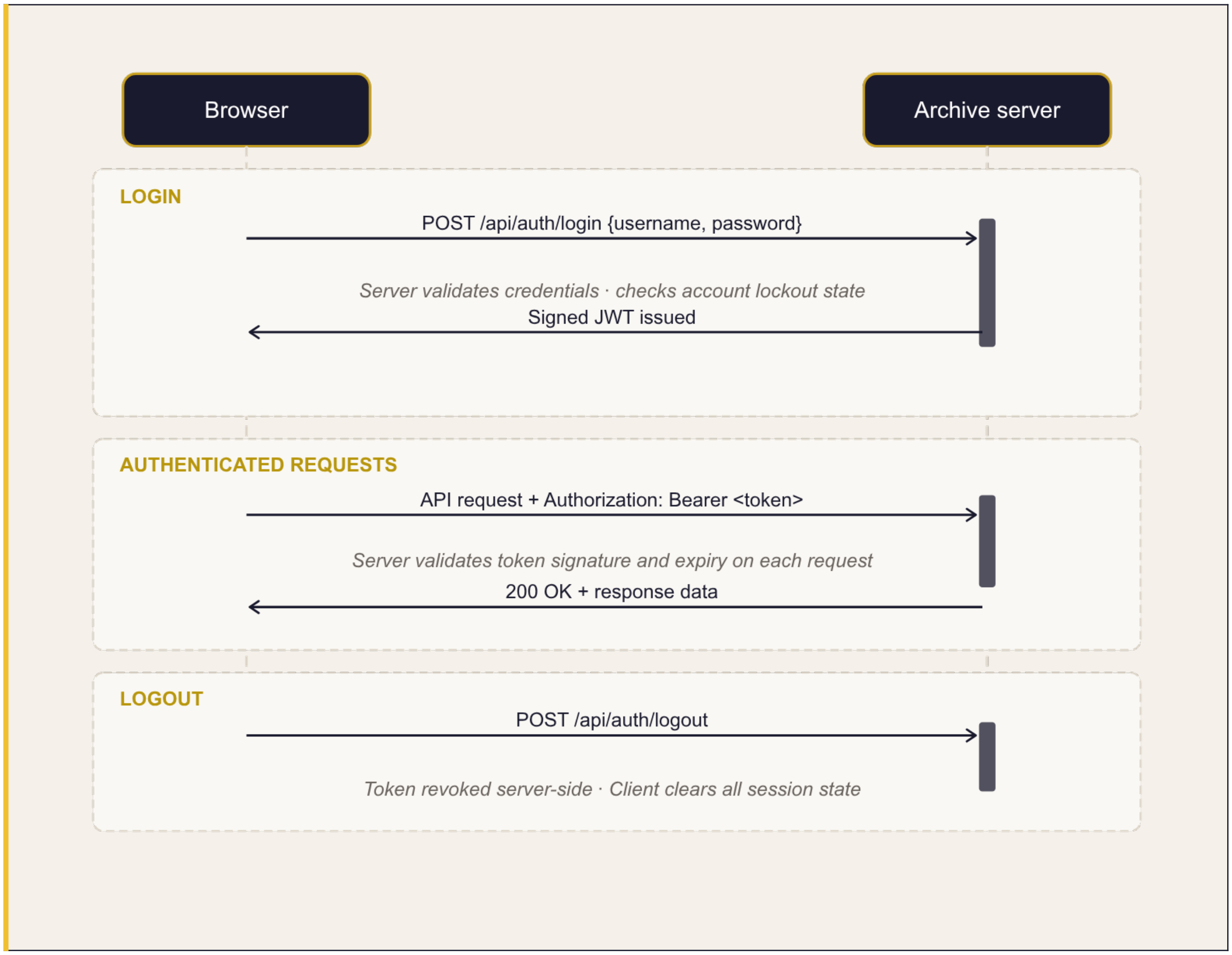
## 11.4 Troubleshooting the Audit Agent

| Symptom | Likely Cause | Resolution |
|---|---|---|
| SSL certificate error | Archive Server uses self-signed cert. | Set verify_ssl = false in config.ini, or install the archive CA certificate into the Windows trust store. |
| API key rejected (401) | Key was revoked or copied incorrectly. | Generate a new API key in the archive and update api_key in config.ini. |
| Audit log path error | server_audit.log not found at configured path. | Locate the correct path in the BigFix installation directory and update audit_log_path. |
| Connection refused / timeout | Archive Server unreachable on port 443. | Verify the firewall allows the BigFix server to reach the archive on port 443. |
| No new records submitted | Position file is ahead of the current log. | Delete position.json to force a full re-read of the audit log from the beginning. |

# 12. Security Reference

## 12.1 Authentication Model

Action Archive uses local authentication with JSON Web Tokens (JWT). There is no Active Directory or LDAP integration in this version.




Tokens are revoked server-side on logout. A revoked token list is maintained in the database for the duration of the token's original validity period, preventing token reuse.

## 12.2 Password Self-Service (Forgot Password)

When SMTP is configured, users can reset their own password using the Forgot Password link on the login page. Reset tokens are stored as SHA-256 hashes; the plaintext appears only in the email link. Tokens expire after 15 minutes and are invalidated immediately after a single use.

Self-service reset also depends on the Password Reset Requests switch under Administration > System Settings > Notifications. If an administrator disables that switch, the Forgot Password flow stops sending reset emails, and passwords must then be reset by an administrator from the User Management page.

## 12.3 Credential Encryption

Sensitive values in the application configuration are encrypted at rest using a machine-derived key that is not written to disk. Encrypted values include: the BigFix operator password, PostgreSQL connection URL, application secret key, and SMTP password.

> ❗ **IMPORTANT** Encrypted credentials are machine-specific. If the application is migrated to a different server, all credentials must be re-entered through the web interface or the installer, as values encrypted on the original machine cannot be decrypted elsewhere.

## 12.4 Infrastructure Security Summary

| Security Control | Implementation |
|---|---|
| Transport Security | TLS 1.2 and 1.3 only. Weak ciphers disabled. HSTS header enforced. |
| Reverse Proxy Isolation | Nginx terminates all TLS. FastAPI binds to localhost:5000 only and is not reachable externally. |
| Security Headers | HSTS, X-Frame-Options: DENY, X-Content-Type-Options: nosniff, X-XSS-Protection, Referrer-Policy. |
| CORS Policy | Cross-origin requests are disabled. The API only accepts requests from the same origin. |
| API Documentation | Swagger UI, ReDoc, and OpenAPI schema endpoints are disabled in production builds. |
| File Permissions | .env and manifest files are mode 600 (owner read/write only). Routers are mode 640. |
| Bytecode Prevention | PYTHONDONTWRITEBYTECODE=1 prevents compiled .pyc files from being written to disk. |
| Rate Limiting | IP-based login lockout after configurable failed attempts (Section 9.2). |
| Audit Trail | All user actions logged with timestamp, source IP, and user agent (Section 10.1). |

## 12.5 Integrity Verification

Action Archive maintains a manifest of expected cryptographic hashes for all application files (.manifest.json). At startup and before each collection, the application verifies that no files have been modified. If the check fails:

- Further collection runs are blocked.
- An integrity failure flag is shown on the About panel.
- The X-Content-Signature: FAILED header is included in API responses to alert monitoring tools.

> ⚠️ **WARNING** A failed integrity check may indicate an incomplete update or unauthorized file modification. Review application logs and compare file hashes manually before re-enabling collections.

# 13. Service Management Reference

## 13.1 Linux Service Commands

The application runs as a systemd service named bigfix-archive. Run the following from a root shell or with sudo:

```
# Check service status
sudo systemctl status bigfix-archive

# Start / stop / restart
sudo systemctl start bigfix-archive
sudo systemctl stop bigfix-archive
sudo systemctl restart bigfix-archive   # required after .env edits

# Enable auto-start on server boot
sudo systemctl enable bigfix-archive

# View recent application logs (last 100 lines)
sudo journalctl -u bigfix-archive -n 100

# Follow application logs in real time
sudo journalctl -u bigfix-archive -f

# Reload Nginx after manual certificate changes
sudo systemctl reload nginx
```

## 13.2 Key File Locations

| Item | Path |
|---|---|
| Application Directory | /opt/bigfix-archive/ |
| Configuration File | /opt/bigfix-archive/config/.env |
| Python Virtual Env | /opt/bigfix-archive/venv/ |
| Static Frontend Files | /opt/bigfix-archive/static/ |
| Application Manifest | /opt/bigfix-archive/.manifest.json |
| Backup Directory | /opt/bigfix-archive/backups/ |
| SSL Certificate | /etc/ssl/certs/bigfix-archive.crt |
| SSL Private Key | /etc/ssl/private/bigfix-archive.key |
| Nginx Site Config | /etc/nginx/sites-available/bigfix-archive |

| Item | Path |
|---|---|
| Systemd Service Unit | /etc/systemd/system/bigfix-archive.service |
| Application Logs | journalctl -u bigfix-archive (no physical log file) |
| Nginx Access Log | /var/log/nginx/bigfix-archive-access.log |
| Nginx Error Log | /var/log/nginx/bigfix-archive-error.log |
| PostgreSQL Database | bigfix_archive (manage via psql or pgAdmin) |

## 13.3 Changing Configuration

Most settings are managed through the web interface and take effect immediately. A service restart is required only when the .env file is edited directly:

```
sudo systemctl restart bigfix-archive
```

> ℹ️ **NOTE** In normal operation you should not need to edit .env directly. Use the web interface for all settings changes. Direct .env edits are reserved for recovery scenarios or initial setup.

# 14. Troubleshooting

The following table covers the most common issues encountered when deploying or operating Action Archive.

| Symptom | Likely Cause | Resolution |
|---|---|---|
| 502 Bad Gateway | FastAPI application not running. | Run: sudo systemctl status bigfix-archive. If stopped, restart. Check journalctl -u bigfix-archive for errors. |
| Login page does not load (connection refused) | Nginx not running. | Run: sudo systemctl status nginx. Restart if stopped. |
| Collection returns 0 actions (expected number of actions to be collected is >0) | BigFix operator lacks Master Operator privileges. | Log in to the BigFix Console, verify the configured operator has the Master Operator role assigned. |
| SSL certificate warning in browser | Self-signed certificate in use. | Expected. Proceed through the browser warning. For a permanent fix, replace with a CA-signed cert (Section 9.5). |
| Login fails after correct password | Account locked from failed attempts. | Admin unlocks via User Management, or wait for Lockout Duration to expire. |
| "Account temporarily locked" message | Rate limiter triggered. | Wait for lockout duration. Consider increasing the attempt threshold if users are frequently locked. |
| Cannot reach BigFix server | Firewall rule or routing issue. | Test: curl -sk https://<bigfix-server>:52311. Check firewall rules between the archive and BigFix Root Server. |
| Password reset email not received | SMTP not configured or misconfigured. | Check Settings → Email. Use Send Test Email to verify. Also check recipient spam folder. |
| Collection is slow on first run | Large number of actions fetched for the first time. | Expected. First run may take several minutes. Subsequent runs are incremental and significantly faster. |
| Audit Agent "connection refused" | Archive URL incorrect or port 443 blocked. | Verify api_url in config.ini. Test: curl -sk https://<archive-server> from the BigFix server. |
| Audit Agent "API key rejected" (401) | Key revoked or incorrectly copied. | Generate a new API key in the archive and update api_key in config.ini. |
| Integrity check failed (About page) | Application file modified or corrupted. | Review application logs, check for unauthorized changes, or reinstall from a known-good package. |
| Database size growing unexpectedly | Pruning not configured. | Enable scheduled pruning in Administration → Settings → Maintenance. |
| Backup download fails or is empty | Disk space exhausted on /opt partition. | Check: df -h. Delete old backup files in Administration → Settings → Backup & Restore. |

# Appendix A: API Reference

The complete REST API specification is distributed as an OpenAPI 3.x document. The file can be imported into tools such as Swagger Editor (editor.swagger.io), Postman, Insomnia, or openapi-generator to browse the full set of endpoints, request and response schemas, and status codes, or to generate client SDKs in different languages.

Location:

• On the Archive Server after installation: /opt/bigfix-archive/docs/openapi.json

• In the public GitHub repository: docs/openapi.json

Authentication

Most endpoints require authentication. Two schemes are supported:

• Bearer JWT, sent as Authorization: Bearer <token>. Obtain a token from POST /api/auth/login. Used by the Archive Server's web UI and recommended for all programmatic integration.

• X-API-Key header, exclusive to the Audit Agent endpoints (GET /api/admin/deleted-action-ids and POST /api/admin/deleted-actions). Keys are created and managed under Administration → System Settings → API Keys.

The following endpoints do not require authentication:

- /api/health
- /api/auth/login
- /api/auth/login-config
- /api/auth/password-policy
- /api/auth/forgot-password
- /api/auth/reset-password.

> ℹ️ **NOTE** Standard HTTP status codes apply: 200 Success, 401 Unauthenticated, 403 Insufficient role, 404 Not found, 422 Validation error, 500 Server error.

# Appendix B: Data Storage Reference

The following table describes the categories of data stored by Action Archive at a conceptual level.

| Data Category | What Is Stored |
|---|---|
| Actions | Action metadata, scripts, relevance expressions, issuer, status, dates, and fixlet references |
| Components | Child actions within Multi-Action Groups (MAGs) and Baselines, linked to their parent action |
| Targets | Per-computer status results for each action, including last report time and result messages |
| Users | User accounts, hashed passwords, roles, data-visibility scopes, and lockout state |
| Audit Trail | All user actions with timestamps, source IP addresses, user agents, and event details |
| Collection Runs | Collection history including start time, duration, status, and per-phase log entries |
| Deleted Actions | Action deletion events submitted by the Audit Agent before archival could occur |
| Configuration | All system settings stored as key-value pairs, managed through the web interface |
| API Keys | Hashed API key records for machine-to-machine authentication, with usage tracking |

Database access is restricted to the application service account. Direct database connectivity from client machines is not permitted. For details on the backup and restore process, see Section 9.9.

# Appendix C: Performance Tuning

Action Archive ships with PostgreSQL's stock configuration, which is intentionally conservative so it can start on the smallest of hosts. Sites running on the recommended hardware baseline (4 vCPU, 8 GB RAM, SSD-backed storage, Ubuntu Server 24.04) can safely apply the optional tuning in this appendix to improve query and export performance, particularly as archives grow into the millions of rows. From v1.1.3 the installer detects your server's RAM, CPU, and storage type during setup and offers to apply these tunings automatically. This appendix documents the same values for review or for manual application after install.

> ℹ️ **NOTE** These changes are optional. The default configuration will work on most archives. The difference will be the collection times. Apply this tuning only if you are seeing slow filtered searches, slow CSV exports, or noticeable lag when browsing very large month partitions.

## C.1 Why Tune?

PostgreSQL's defaults assume a tiny server (≈ 128 MB allocated to its own cache) and a spinning hard disk. Action Archive on Ubuntu is not designed to be co-located with the BigFix Root Server, so the database host belongs to PostgreSQL, the Action Archive API, and Nginx alone. That dedicated footprint and the SSD-backed storage assumption in the hardware baseline are the foundation for every value in this appendix.

## C.2 The Tuning Drop-in File

All tuning is applied via a single drop-in file under PostgreSQL's `conf.d` directory. This approach leaves the package-managed `postgresql.conf` untouched, so the file survives PostgreSQL package upgrades and is trivial to remove if you ever need to roll back.

Create the file at **/etc/postgresql/16/main/conf.d/99-action-archive-tuning.conf** with the contents shown below. The values shown are sized for the recommended 8 GB host; consult the sizing table in section C.3 if your server has more or less RAM.

**FILE: /etc/postgresql/16/main/conf.d/99-action-archive-tuning.conf**

```
# Action Archive: PostgreSQL tuning drop-in
# Sized for a dedicated 4 vCPU / 8 GB / SSD host (the recommended baseline).
# Adjust the RAM-dependent values per the sizing table in the Admin Guide.

# --- RAM-dependent (see sizing table) ---
shared_buffers           = 2GB
effective_cache_size     = 5GB
work_mem                 = 16MB
maintenance_work_mem     = 256MB

# --- Fixed: SSD storage assumption ---
random_page_cost         = 1.1
effective_io_concurrency = 200

# --- Fixed: parallelism for a 4-core host ---
max_worker_processes             = 4
```

FILE: /etc/postgresql/16/main/conf.d/99-action-archive-tuning.conf

```
max_parallel_workers            = 4
max_parallel_workers_per_gather  = 2
max_parallel_maintenance_workers = 2

# --- Fixed: WAL and checkpoint behaviour ---
wal_buffers                  = 16MB
checkpoint_completion_target = 0.9
```

## C.3 Sizing Table (Adjust by Server RAM)

Four settings should be sized to match the host's total RAM. The other settings in the drop-in file are fixed and do not depend on memory. Pick the row in the table below that matches your server's installed RAM and edit the four values in the drop-in file accordingly.

| Total RAM | shared_buffers | effective_cache_size | work_mem | maintenance_work_mem |
|---|---|---|---|---|
| 4 GB (minimum) | 512 MB | 2 GB | 8 MB | 128 MB |
| 8 GB (recommended) | 2 GB | 5 GB | 16 MB | 256 MB |
| 16 GB (large archives) | 4 GB | 10 GB | 32 MB | 512 MB |
| 32 GB + (very large) | 8 GB | 22 GB | 64 MB | 1 GB |

**Rationale.** `shared_buffers` is roughly 25 % of total RAM (the page cache PostgreSQL owns directly). `effective_cache_size` is a planner hint at 60 % to 70 % of total RAM (it is not an allocation). `work_mem` is per sort or hash operation inside a query, so the safe ceiling depends on how many sorts can run concurrently. With Action Archive's single Uvicorn worker and small SQLAlchemy connection pool, peak concurrency is low. `maintenance_work_mem` speeds up VACUUM, index builds, and the monthly partition creation job.

## C.4 CPU Parallelism

The drop-in file assumes a 4-core host. If your server has 8 cores (or more), raise the four parallelism settings so the planner can fan large scans across more workers (particularly useful when filtering or exporting across many monthly partitions of `action_targets` or `action_target_history`).

OVERRIDE FOR 8-CORE HOSTS

```
max_worker_processes             = 8
max_parallel_workers             = 8
max_parallel_workers_per_gather  = 4
max_parallel_maintenance_workers = 4
```

## C.5 Kernel Tuning

One Linux kernel setting is worth changing on a database host. Ubuntu's default `vm.swappiness` of 60 causes the kernel to swap database pages out of RAM under modest memory pressure, which defeats the purpose of the larger `shared_buffers` value above. Lower it to 10 so PostgreSQL's hot pages stay resident.

Create the file at **/etc/sysctl.d/99-action-archive.conf**:

FILE: /etc/sysctl.d/99-action-archive.conf

```
# Action Archive: kernel tuning for a database host
vm.swappiness = 10
```

Apply with:

COMMAND

```
sudo sysctl --system
```

## C.6 What Is Deliberately Left Alone

Several PostgreSQL settings could in principle be tuned but provide no measurable benefit at Action Archive's scale and access patterns. They are listed here so administrators familiar with PostgreSQL tuning guides do not wonder why they are absent from the drop-in file.

- **jit:** left enabled at default. JIT only activates above a high cost threshold; harmless at this scale and begins to pay off as archives grow into the tens of millions of rows.
- **default_statistics_target:** left at 100. Worth revisiting only if a single table exceeds roughly 10 million rows.
- **huge_pages:** left at `try`. Only meaningful when `shared_buffers` is 8 GB or larger, and the kernel-side configuration to back it adds operational complexity that is not justified for typical installs.
- **max_connections:** left at 100. Action Archive's peak concurrent backend connections are well under 30.

## C.7 Applying and Verifying the Changes

Several of the values above (`shared_buffers`, `wal_buffers`, `max_worker_processes`) require a full PostgreSQL restart, so restart the service once after creating the drop-in file.

COMMAND

```
sudo systemctl restart postgresql@16-main
sudo systemctl status  postgresql@16-main --no-pager | head
```

Confirm the new values are in effect with a quick `SHOW` from `psql`:

COMMAND

```
sudo -u postgres psql -c "SELECT name, setting, unit, source
                          FROM pg_settings
                          WHERE name IN ('shared_buffers','effective_cache_size',
                                         'work_mem','maintenance_work_mem',

'random_page_cost','effective_io_concurrency',
```

COMMAND

```
'max_worker_processes','max_parallel_workers',

'max_parallel_workers_per_gather','wal_buffers')
                          ORDER BY name;"
```

Every row in the result should show `source = configuration file` (rather than `default`). If any row still says `default`, check the drop-in file's path and permissions, and check the PostgreSQL log for an include-directory parse error.

## C.8 Rollback

If the tuned configuration ever needs to be reversed, remove the drop-in file and restart PostgreSQL. No other state needs to be undone.

COMMAND

```
sudo rm /etc/postgresql/16/main/conf.d/99-action-archive-tuning.conf
sudo rm /etc/sysctl.d/99-action-archive.conf
sudo sysctl --system
sudo systemctl restart postgresql@16-main
```

# Appendix D: Glossary

| Term | Definition |
|---|---|
| Action | A deployment task created in BigFix that runs a script or applies a fixlet to one or more target devices. |
| Action Archive | This application. A web-based system for collecting, preserving, and querying stopped and expired BigFix action records. |
| Audit Agent | The Windows Scheduled Task component that monitors the BigFix server_audit.log and reports action deletions to the archive. |
| Baseline | A grouped BigFix action that deploys multiple fixlets as a single coordinated operation. Stored as a Multi-Action Group. |
| Bearer Token | A JWT credential included in the Authorization HTTP header to authenticate API requests. |
| BigFix Root Server | The central IBM BigFix server that manages policies, actions, and inventory for all managed endpoints. |
| Collection | A scheduled or manual process in which Action Archive queries BigFix for new expired / stopped actions and stores them. |
| Expired Action | A BigFix action whose end date or time has passed. Action Archive archives these actions when a collection runs. |
| Fixlet | A BigFix content item that defines a condition and an action to remediate it. Actions reference fixlets as their source. |
| Fixlet Site | A container in BigFix that groups related fixlets and actions (e.g., Patches for Windows, Custom Tasks). |
| Issuer | The BigFix operator username that created a given action. |
| JWT | JSON Web Token. A signed, time-limited credential used to authenticate web UI sessions. |
| MAG | Multi-Action Group. A parent BigFix action that coordinates a set of member (component) actions. |
| Master Operator (MO) | A BigFix operator with permissions to see and query all actions across all sites. Required for the archive collector account. |
| pg_dump | A PostgreSQL utility used to create portable database backups. Used by the archive's Backup & Restore feature. |
| Pruning | The process of permanently deleting archived action records older than the configured retention period. |
| Scope | A data-visibility restriction assigned to a user that limits which sites or issuers they can see in the archive. |
| Session Relevance | A BigFix query language for retrieving structured data from the BigFix server, used to supplement the REST API. |
| Stopped Action | A BigFix action manually stopped before its end date. Archived alongside |

| Term | Definition |
| --- | --- |
| | expired actions. |
| TLS | Transport Layer Security. The protocol used to encrypt HTTPS communications between browsers and the archive server. |